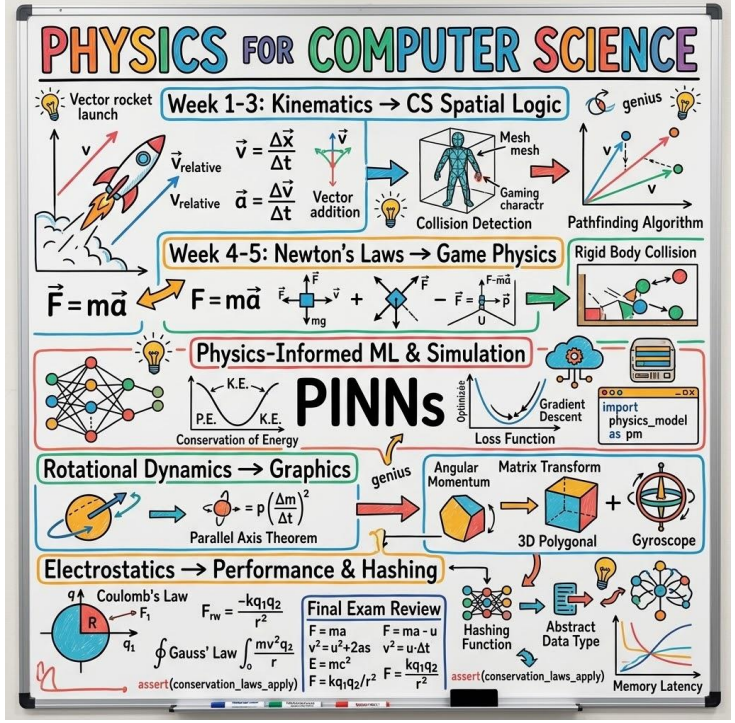
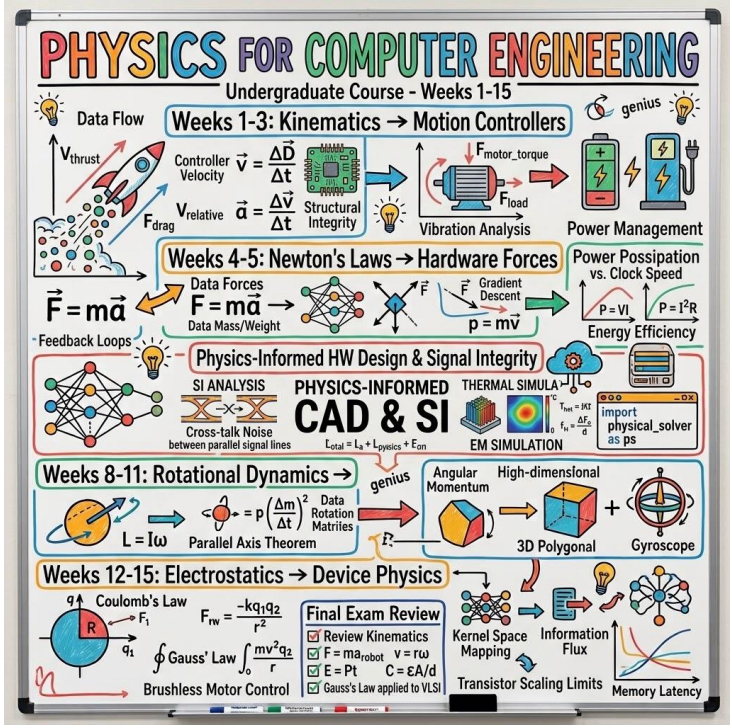
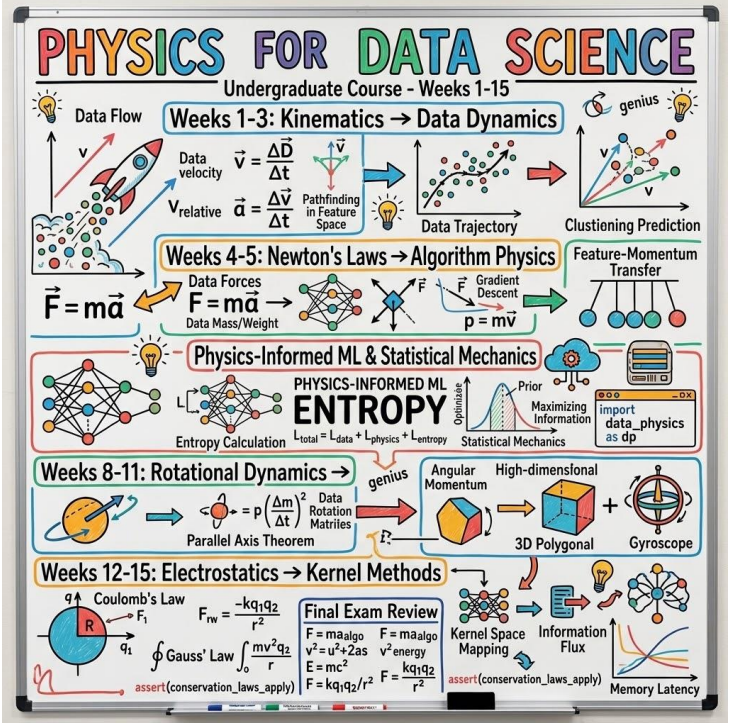
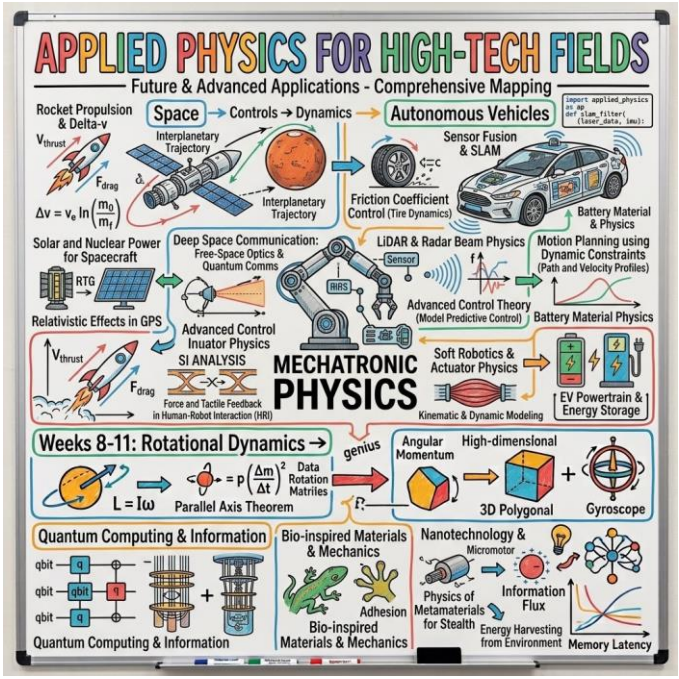








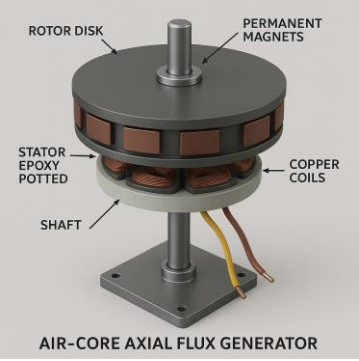
Applied Physics
PH101

Faculty of Basic Sciences

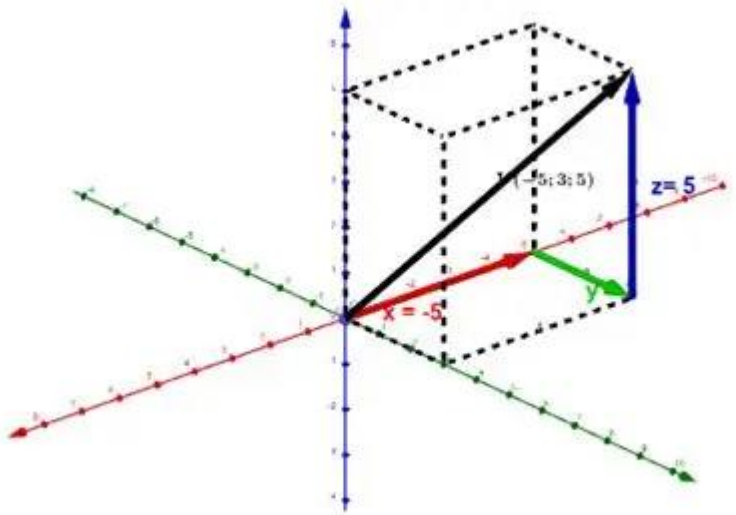
Week_04

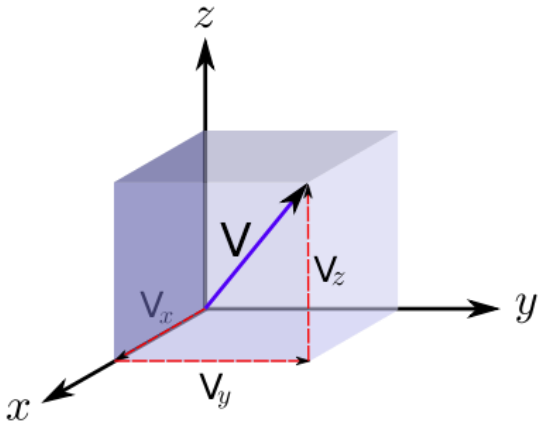
Fall_2026

Weekly Plan



- **Vector's Operations**
- **Numerical Examples**
- **Motion in Two or Three Dimensions**





Components of a Vector

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

A 3D vector **B** has a magnitude of 20 units and makes angles α=60° with x-axis, β=45° with y-axis, and γ=60° with z-axis.

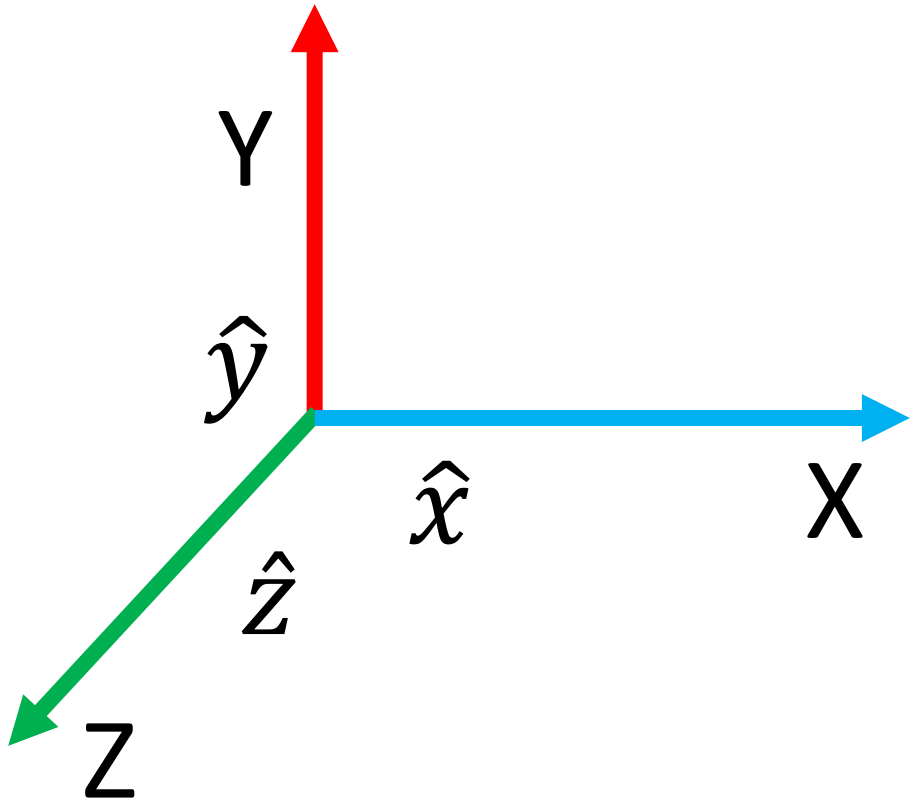
The rectangular components are:

$$B_x = B\cos\alpha = 20\cos 60^\circ = 20 \times 0.5 = 10$$

$$B_y = B\cos\beta = 20\cos 45^\circ = 20 \times 0.707 = 14.14$$

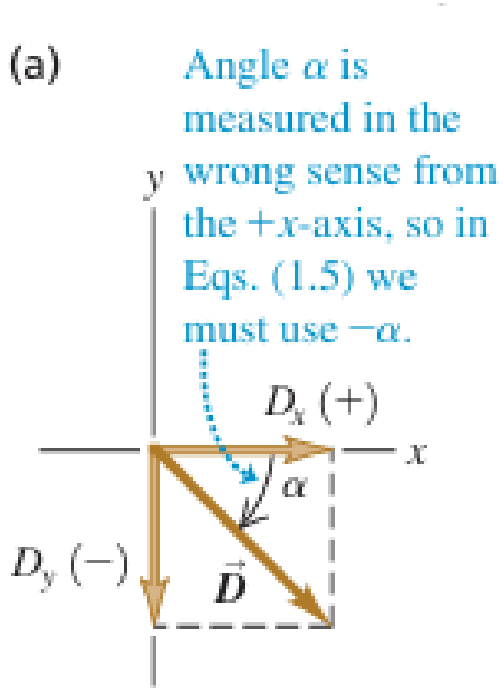
$$B_z = B\cos\gamma = 20\cos 60^\circ = 20 \times 0.5 = 10$$

$$\vec{B} = 10\,\hat{\imath} + 14.14\,\hat{\jmath} + 10\hat{k}$$



Example

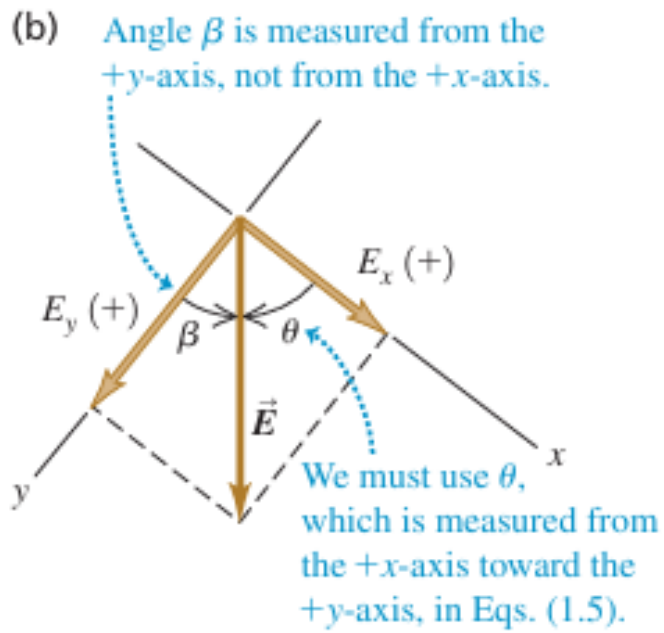
(a) What are the x- and y-components of vector $\vec{D}$ in figure? The magnitude of the vector is $D = 3.00$ m, and angle $\alpha = 45°$



$$D_x = D\cos\theta = (3.00\text{ m})(\cos(-45°)) = +2.1\text{ m}$$
$$D_y = D\sin\theta = (3.00\text{ m})(\sin(-45°)) = -2.1\text{ m}$$

Example

(b) What are the x- and y-components of vector $\vec{E}$ in figure? The magnitude of the vector is $E = 4.50$ m, and angle $\beta = 37.0°$.



(b) The x- and y-axes in figure b, are at right angles, so it doesn't matter that they aren't horizontal and vertical, respectively. But we can't use the angle β(the Greek letter beta), because $\beta$is measured from the $+y$-axis. Instead, we must use the angle $\theta = 90.0° - \beta = 90.0° - 37.0° = 53.0°$.

$$E_x = E\cos 53.0° = (4.50\text{ m})(\cos 53.0°) = +2.71\text{ m}$$
$$E_y = E\sin 53.0° = (4.50\text{ m})(\sin 53.0°) = +3.59\text{ m}$$

Components of a Vector

$$A = \sqrt{A_x^2 + A_y^2}$$

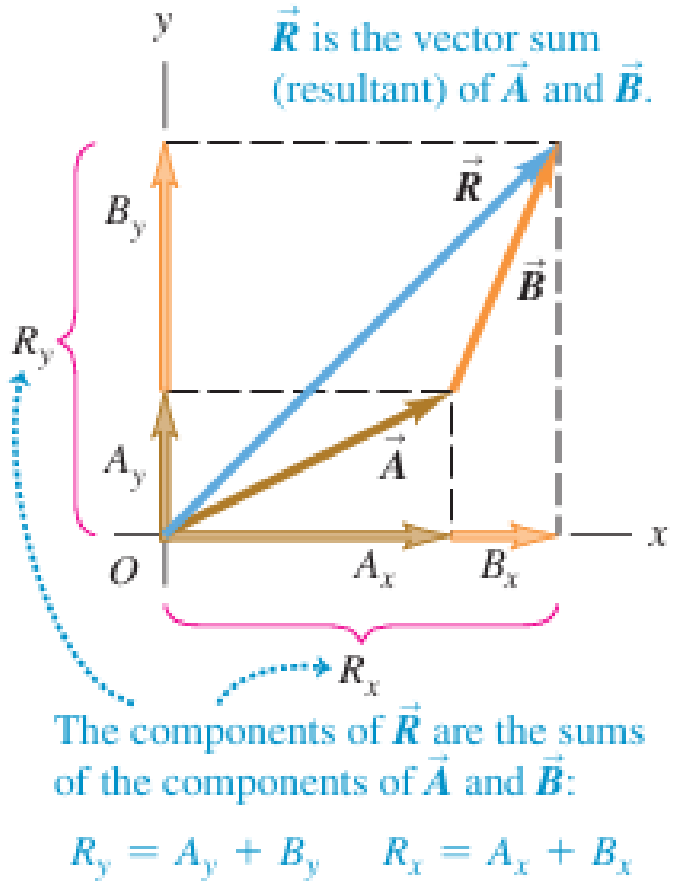
$$\tan\theta = \frac{A_y}{A_x} \qquad \text{and} \qquad \theta = \arctan\frac{A_y}{A_x}$$

Each **component of** $\vec{R} = \vec{A} + \vec{B}$ ...

$$R_x = A_x + B_x, \qquad R_y = A_y + B_y$$

... is the sum of the corresponding components of $\vec{A}$ and $\vec{B}$.

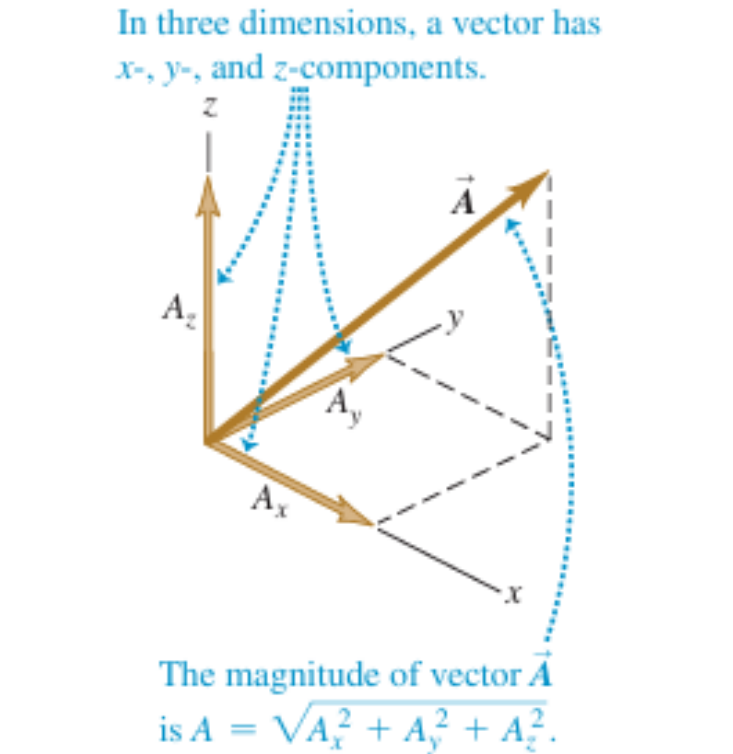
Figure 1.21 Finding the vector sum (resultant) of $\vec{A}$ and $\vec{B}$ using components.



Components of a Vector

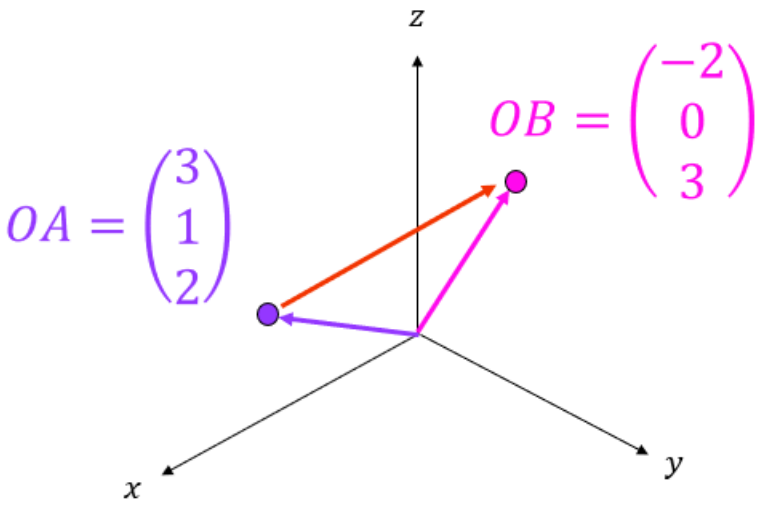
$$R_x = A_x + B_x + C_x + D_x + E_x + \cdots$$
$$R_y = A_y + B_y + C_y + D_y + E_y + \cdots$$
$$R_z = A_z + B_z + C_z + D_z + E_z + \cdots$$

Figure 1.22 A vector in three dimensions.



Example

Find the vector from A(3, 1, 2) to B(-2, 0, 3)



$$\overrightarrow{AB} = \begin{pmatrix} -2-3 \\ 0-1 \\ 3-2 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} -5 \\ -1 \\ 1 \end{pmatrix}$$

Three players on a reality TV show are brought to the center of a large, flat field. Each is given a meter stick, a compass, a calculator, a shovel, and (in a different order for each contestant) the following three displacements:



$\vec{A}$: 72.4 m, 32.0° east of north
$\vec{B}$: 57.3 m, 36.0° south of west
$\vec{C}$: 17.8 m due south

The three displacements lead to the point in the field where the keys to a new Porsche are buried. Two players start measuring immediately, but the winner first *calculates* where to go. What does she calculate?

Example

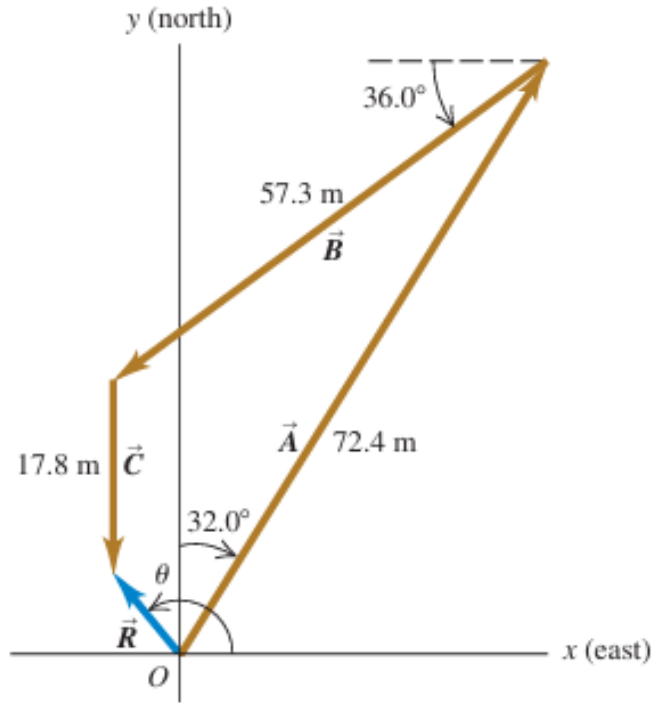
Distance	Angle	x-component	y-component
$A = 72.4$ m	58.0°	38.37 m	61.40 m
$B = 57.3$ m	216.0°	−46.36 m	−33.68 m
$C = 17.8$ m	270.0°	0.00 m	−17.80 m
		$R_x = -7.99$ m	$R_y = 9.92$ m

$$R = \sqrt{(-7.99\ \text{m})^2 + (9.92\ \text{m})^2} = 12.7\ \text{m}$$

$$\theta = \arctan\frac{9.92\ \text{m}}{-7.99\ \text{m}} = -51°$$

The correct value is 180°+(-51°) = 129°, or 39° west of north.

Figure 1.23 Three successive displacements $\vec{A}$, $\vec{B}$, and $\vec{C}$ and the resultant (vector sum) displacement $\vec{R} = \vec{A} + \vec{B} + \vec{C}$.



Example

Find the angle between the vectors

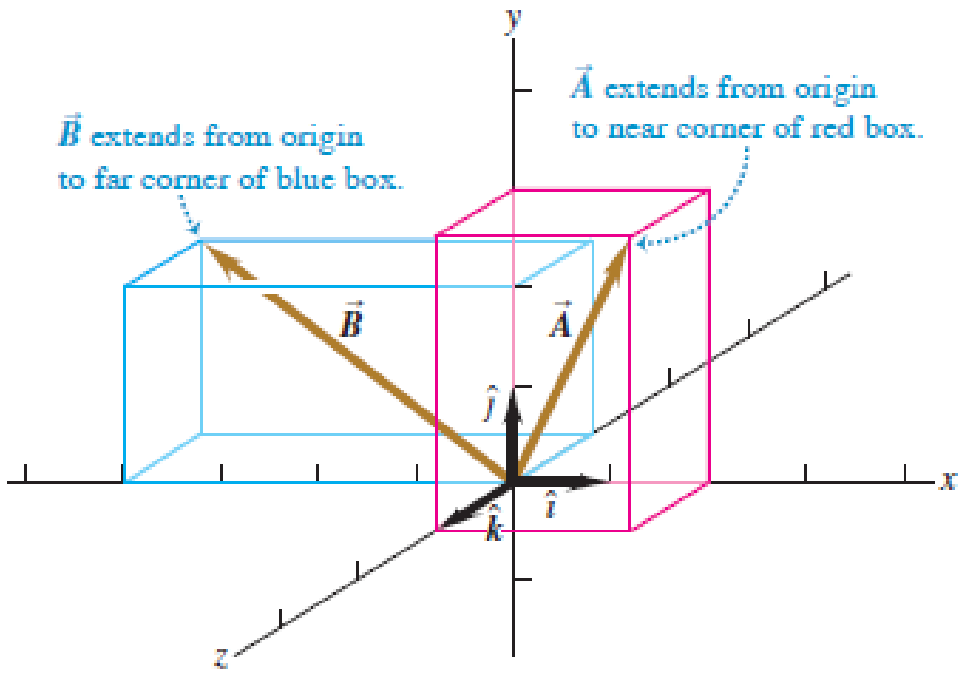
$$\vec{A} = 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k}$$

and

$$\vec{B} = -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}$$

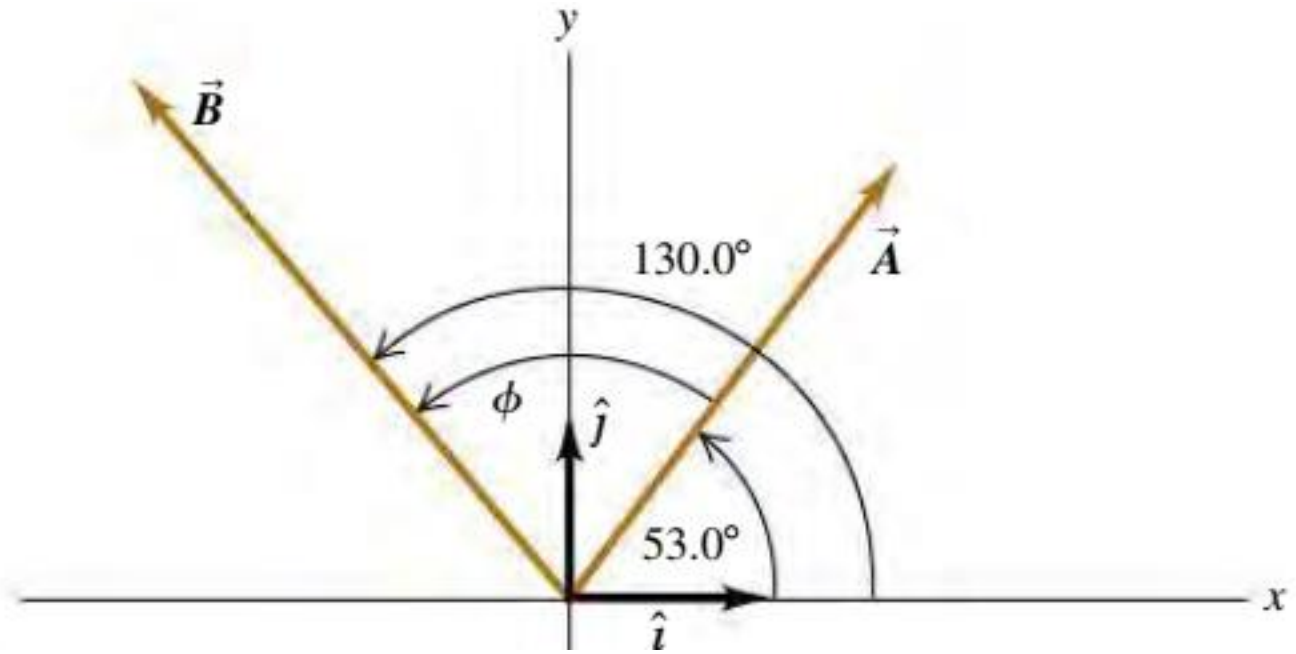
$$\cos\phi = \frac{\vec{A}\cdot\vec{B}}{AB}$$

Figure 1.29 Two vectors in three dimensions.

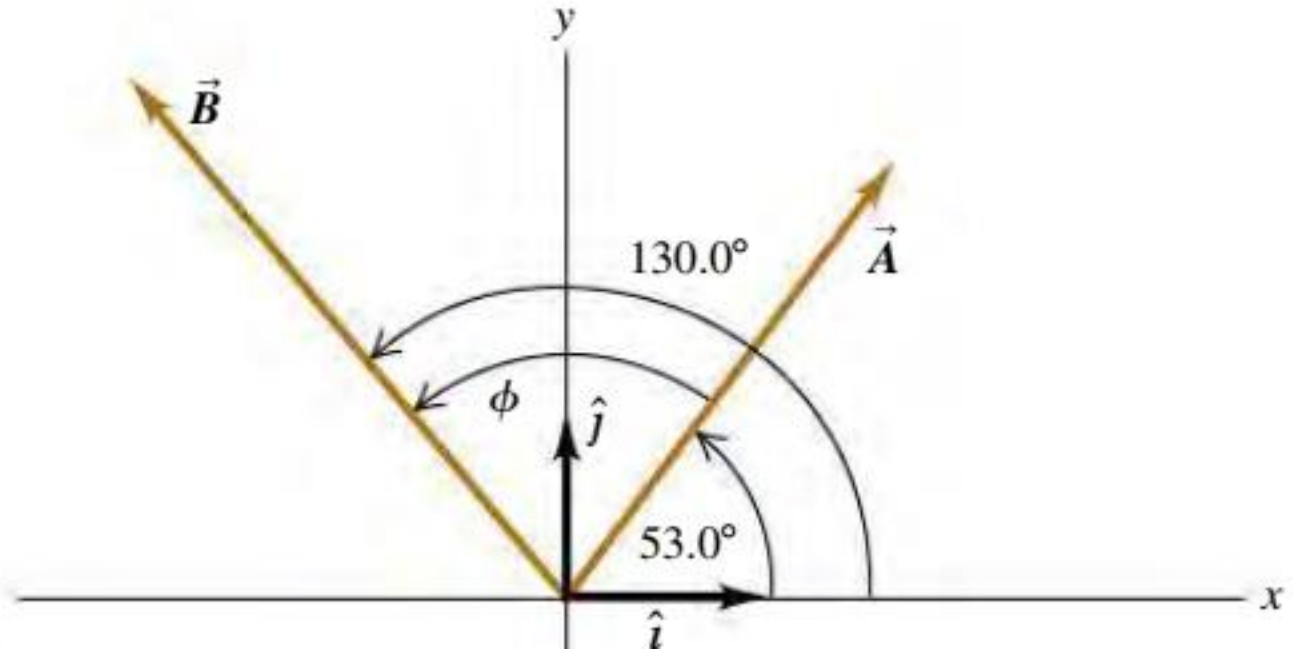


Examples

Find the scalar product $\vec{A}.\vec{B}$ of the two vectors shown in figure. The magnitude of the vector are 4 and 5 respectively.



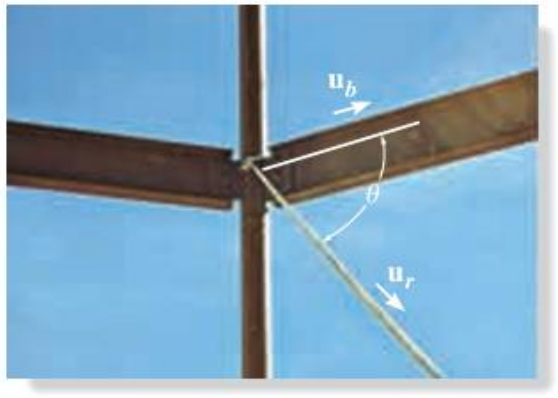
Examples



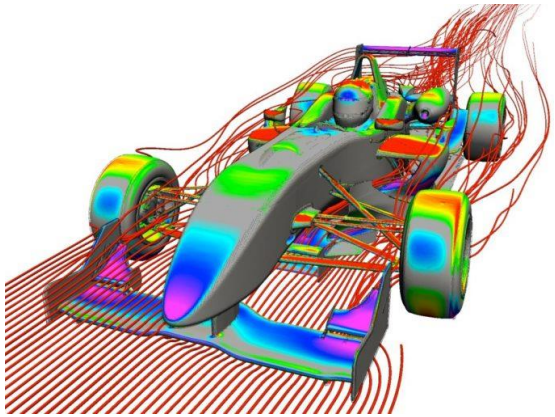
Examples (Vector Multiplication)

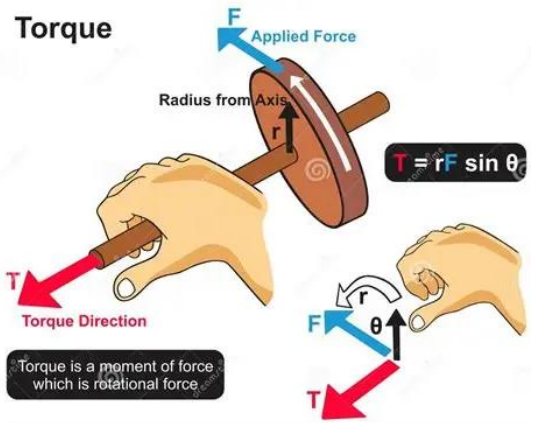
$$Work = \overrightarrow{Displacement} \cdot \overrightarrow{Force}$$

$$Torque = \overrightarrow{Displacement} \times \overrightarrow{Force}$$



The angle θ between the rope and the beam can be determined by formulating unit vectors along the beam and rope and then using the dot product $\mathbf{u}_b \cdot \mathbf{u}_r = (1)(1)\cos\theta$. (© Russell C. Hibbeler)



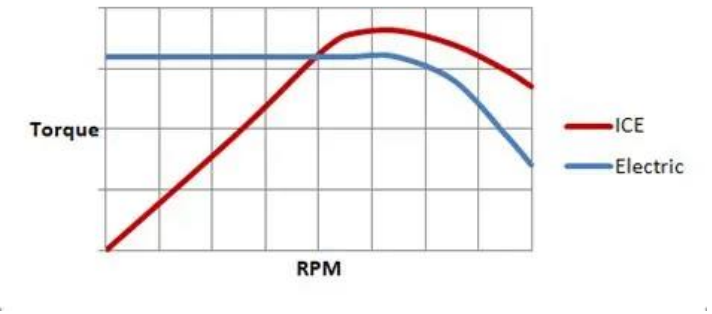


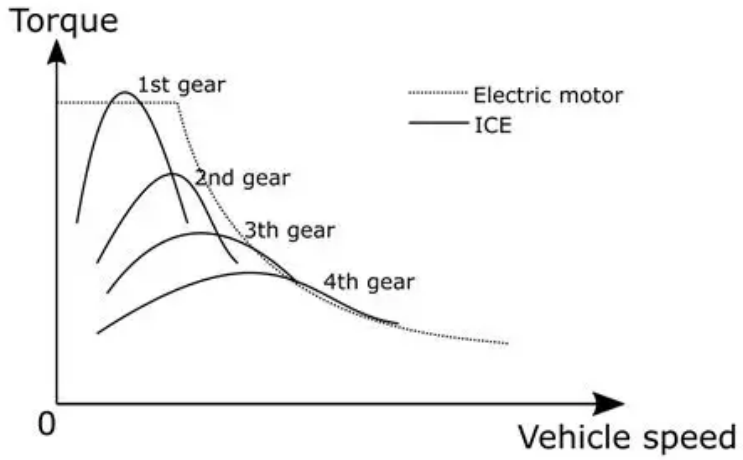


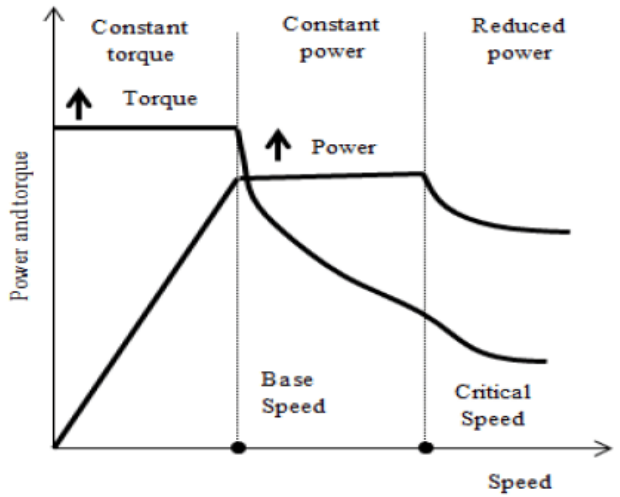
Examples (Vector Multiplication)

$$Work = \overrightarrow{Displacement} \cdot \overrightarrow{Force}$$

$$Torque = \overrightarrow{Displacement} \times \overrightarrow{Force}$$







Example

- Three players on a reality TV show are brought to the center of a large, flat field. Each is given a meter stick, a compass, a calculator, a shovel, and (in a different order for each contestant) the following three displacements:
- ***A***: 72.4 m, 32.0° east of north
- ***B***: 57.3 m, 36.0° south of west
- ***C***: 17.8 m due south
- The three displacements lead to the point in the field where the keys to a new Porsche are buried. Two players start measuring immediately, but the winner first *calculates* where to go. What does she calculate?

Example

$$R_x = A_x + B_x + C_x$$

$$R_y = A_y + B_y + C_y$$

- ***A***: 72.4 m, 32.0° east of north
- ***B***: 57.3 m, 36.0° south of west
- ***C***: 17.8 m due south

$$A_x = 72.4\ \mathrm{Cos}(90 - 32)$$

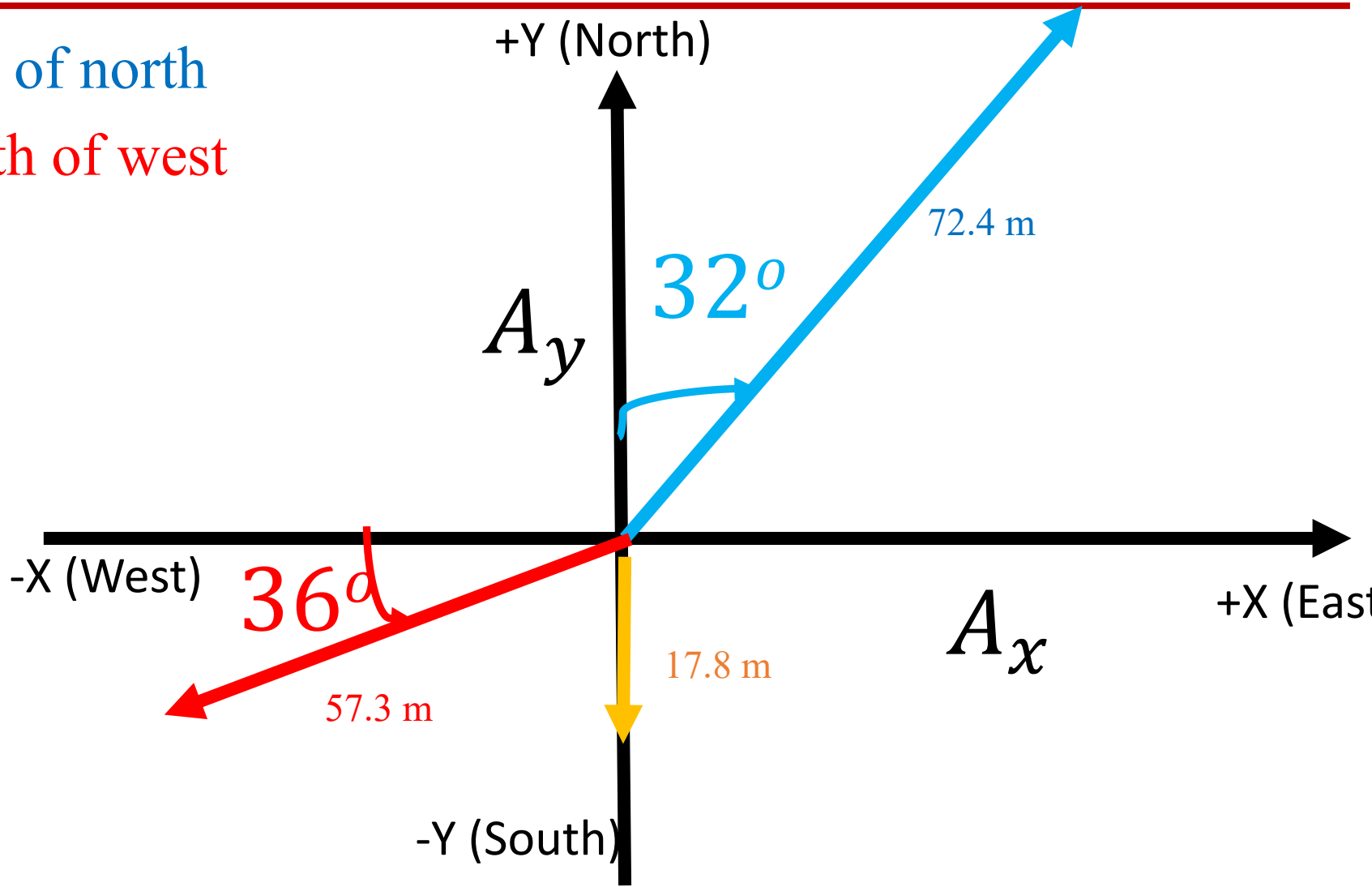
$$A_y = 72.4\ Sin(90 - 32)$$

$$A_x = 57.3\ \mathrm{Cos}(180 + 36)$$

$$A_y = 57.3\ Sin(180 + 36)$$

$$A_x = 17\ \mathrm{Cos}(270)$$

$$A_y = 17\ Sin(270)$$



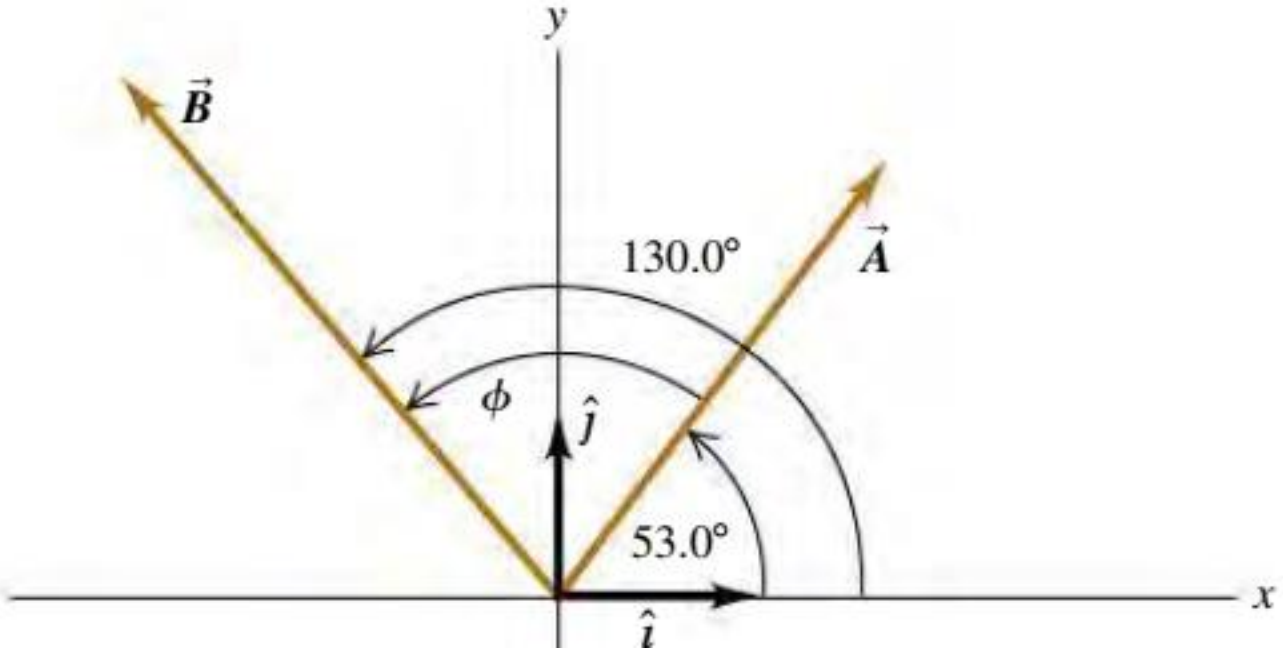
Examples

Find the scalar product $\vec{A}.\vec{B}$ of the two vectors shown in figure. The magnitude of the vector are 4 and 5 respectively.

$\vec{A}.\vec{B}=|A||B|Cos(\theta)$

$\vec{A}.\vec{B} = (4)(5)\,Cos(77)$

$\vec{A}.\vec{B}=4.50$



Examples

Find the scalar product $\vec{A} \cdot \vec{B}$ of the two vectors in **Fig. 1.28.** The magnitudes of the vectors are $A = 4.00$ and $B = 5.00$.

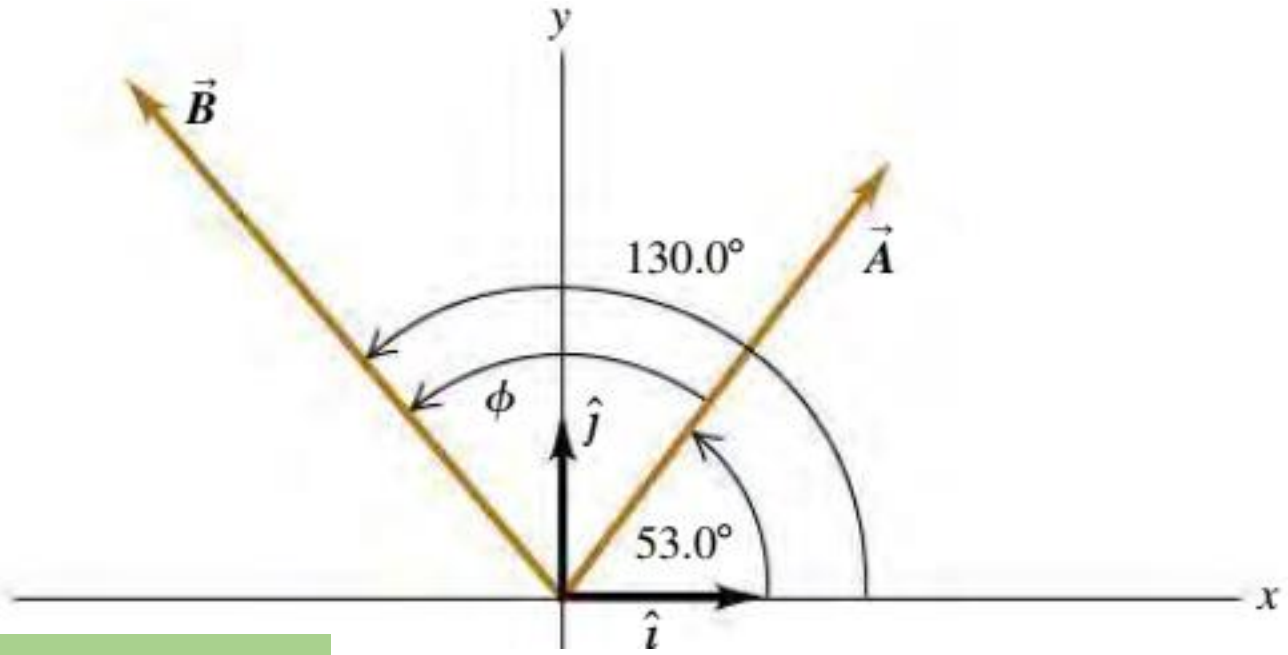
$$A_x = 4\,Cos(53) = 2.407$$

$$A_y = 4\,Sin(53) = 3.195$$

$$B_x = 5\,Cos(130) = -3.214$$

$$B_y = 5\,Sin(130) = 3.830$$

$$\vec{A}.\vec{B} = A_xB_x + A_yB_y + A_zB_z = 4.50$$



Example

Determine the magnitude and direction θ of F_A so that the resultant force is directed along the positive x axis and has a magnitude of 1250 N.

$$R_x = F_A Cos\,(90 - \theta) + 800\,Cos(30)$$

$$R_x = F_A\,Sin(\theta) + 800\,Cos(30) = 1250$$

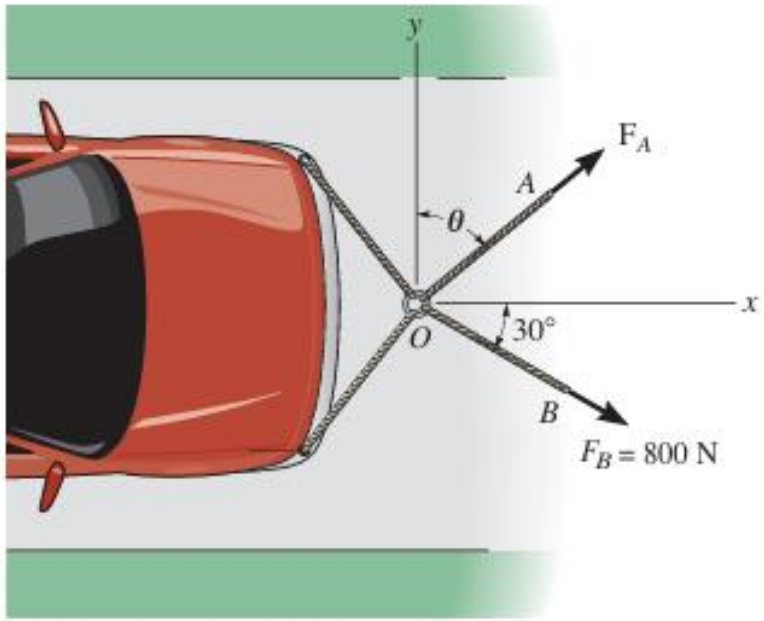
$$R_y = F_A\,Cos\,(\theta) - 800\,Sin(30) = 0$$

$$\theta = ?$$

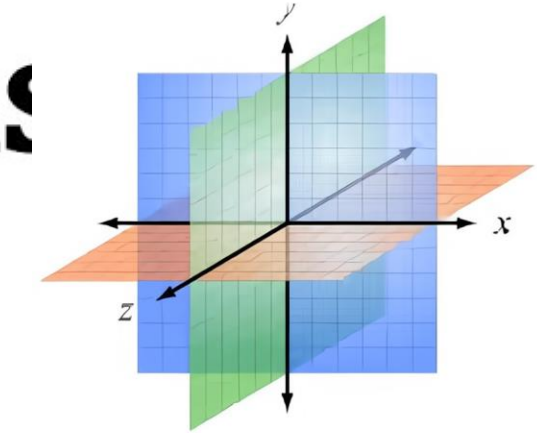
$$F_A = ?$$

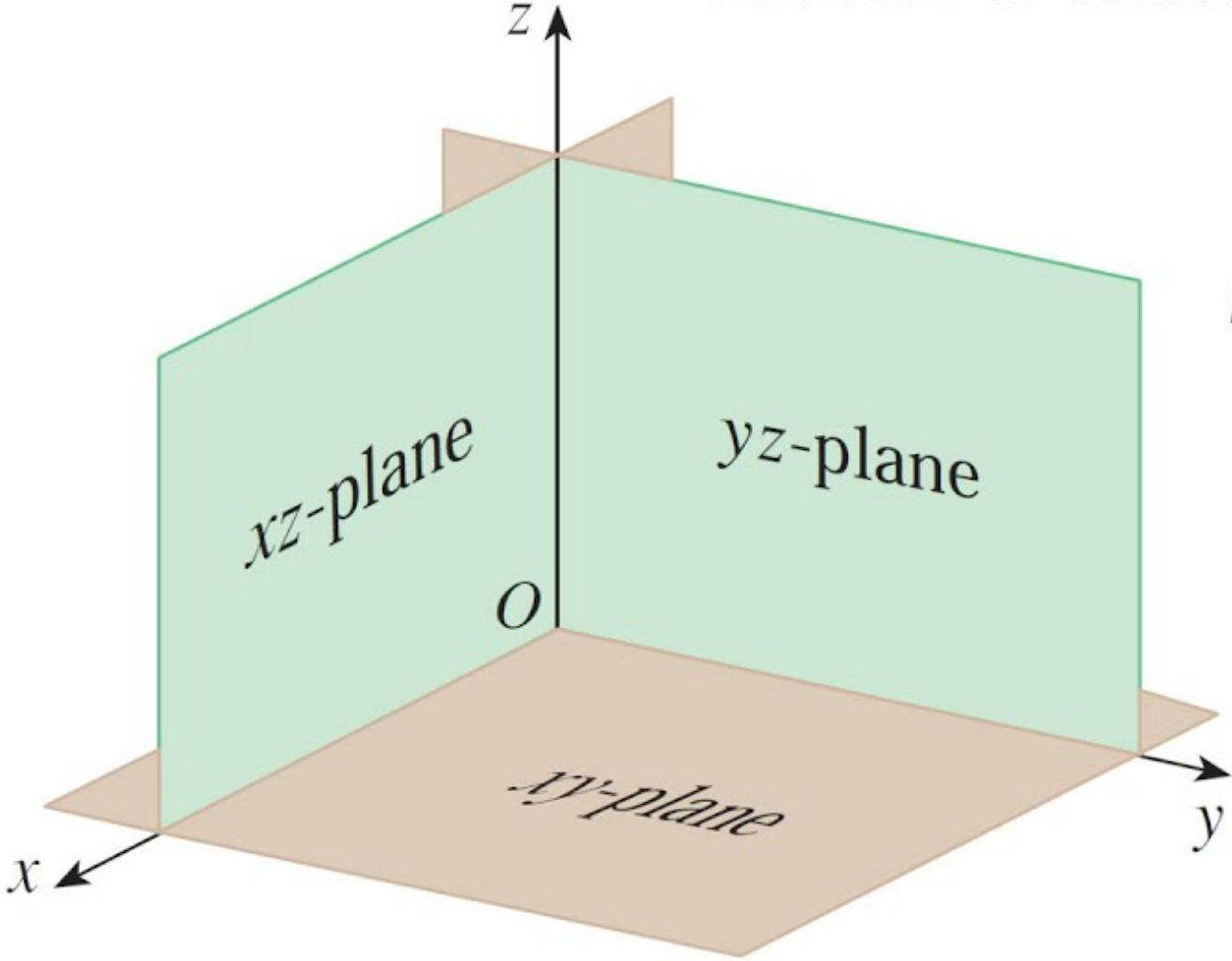
$$\theta = 54.32$$

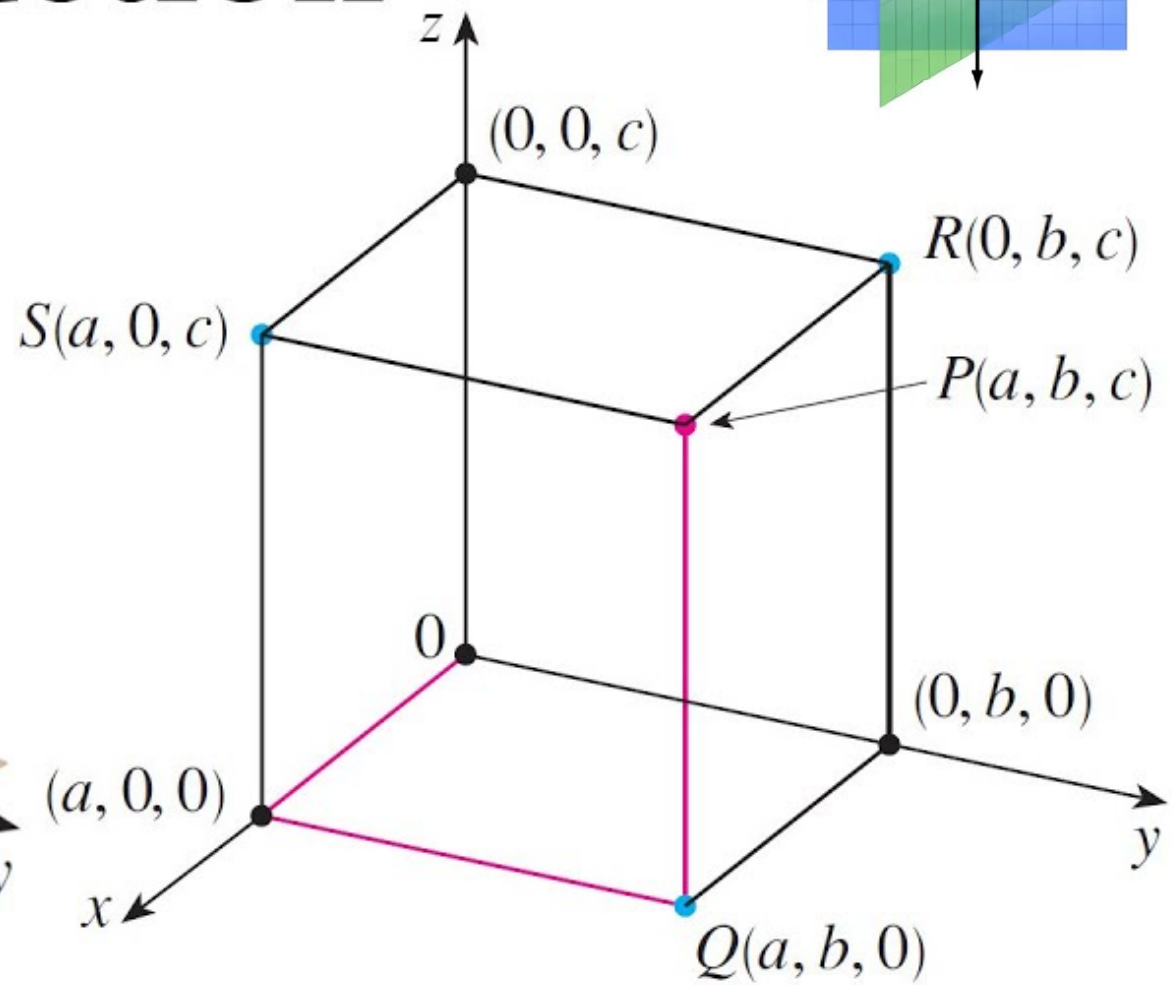
$$F_A = 585.91\,N$$

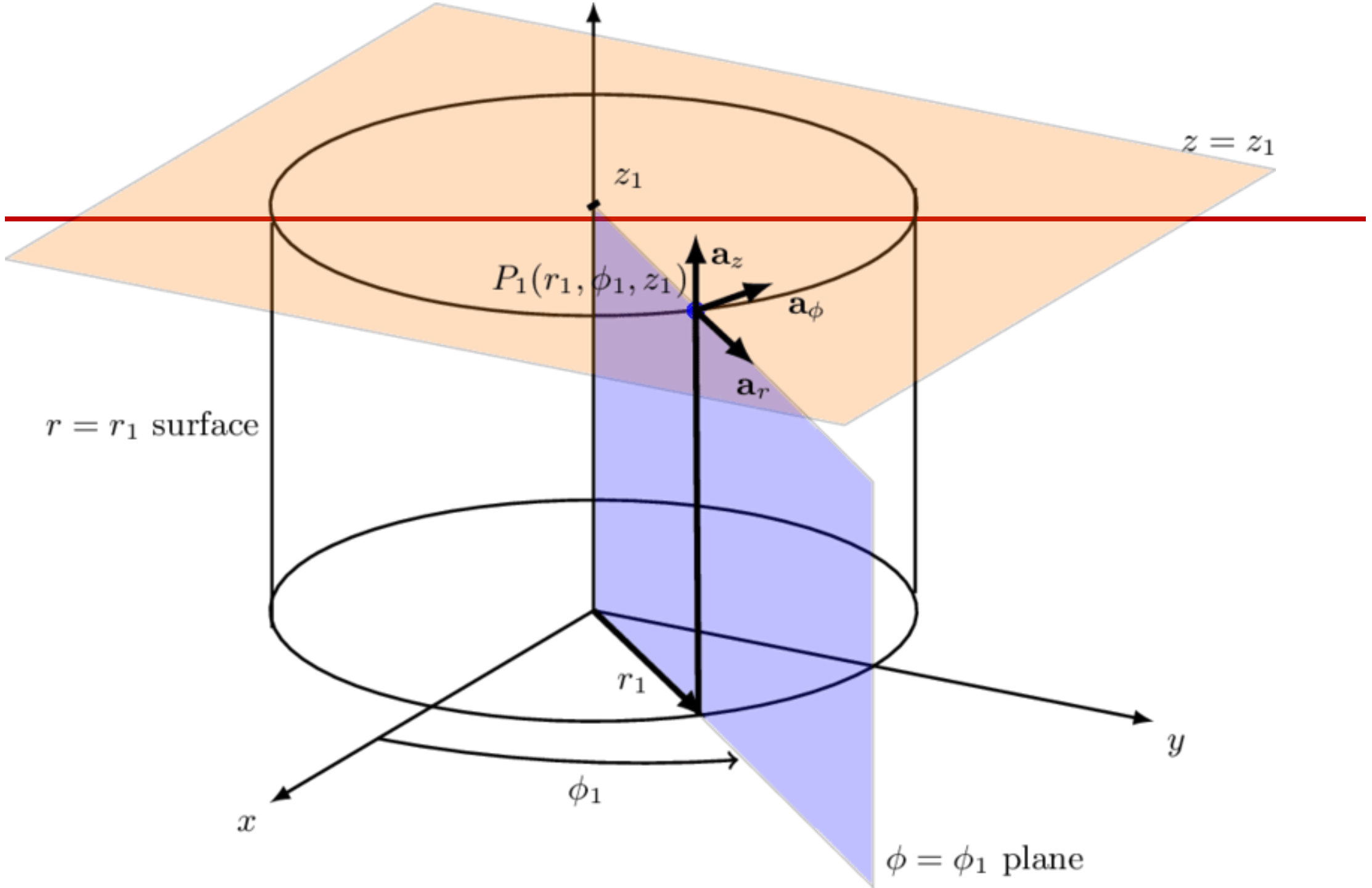


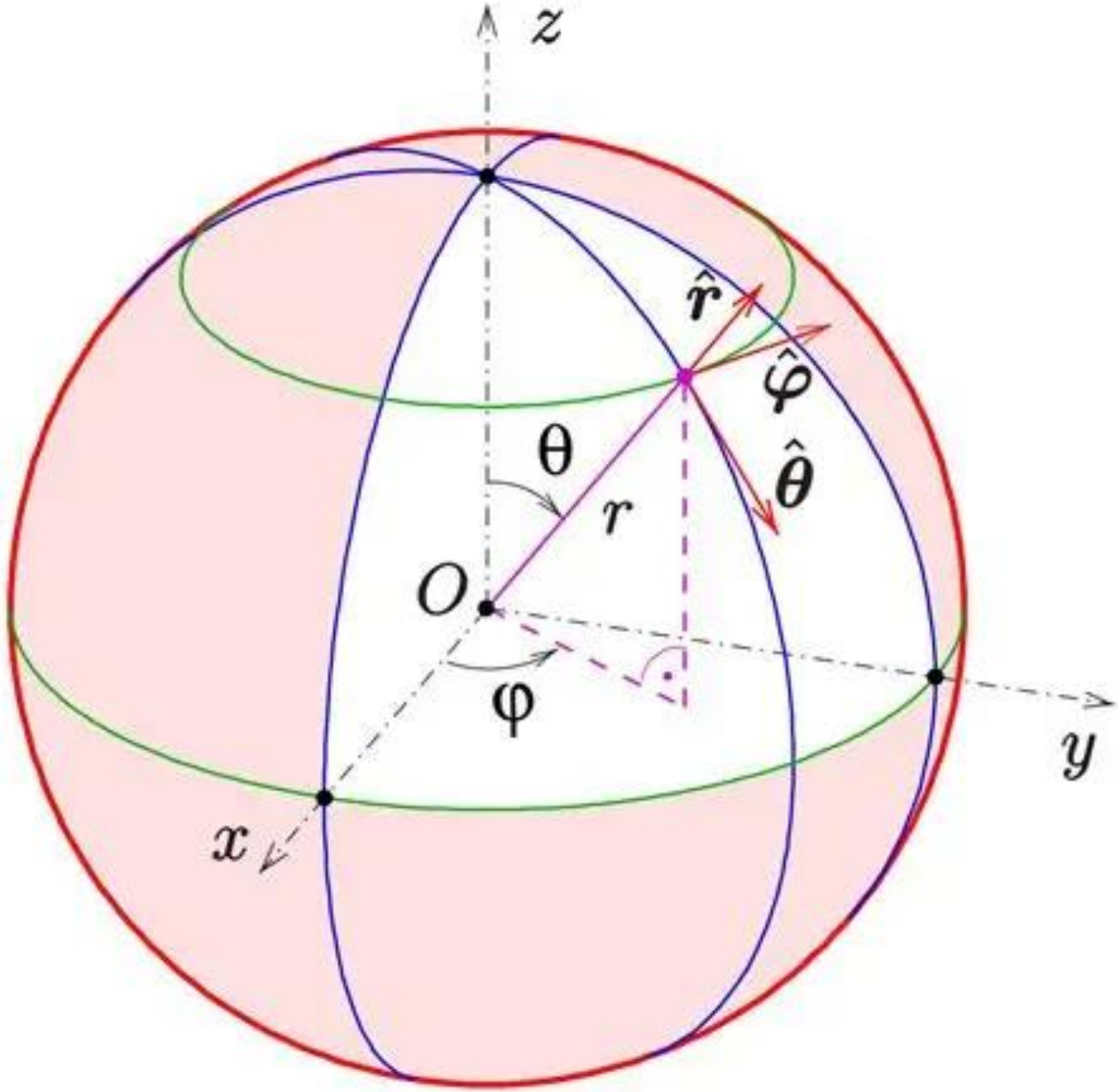
3D Coordinate Systems Introduction



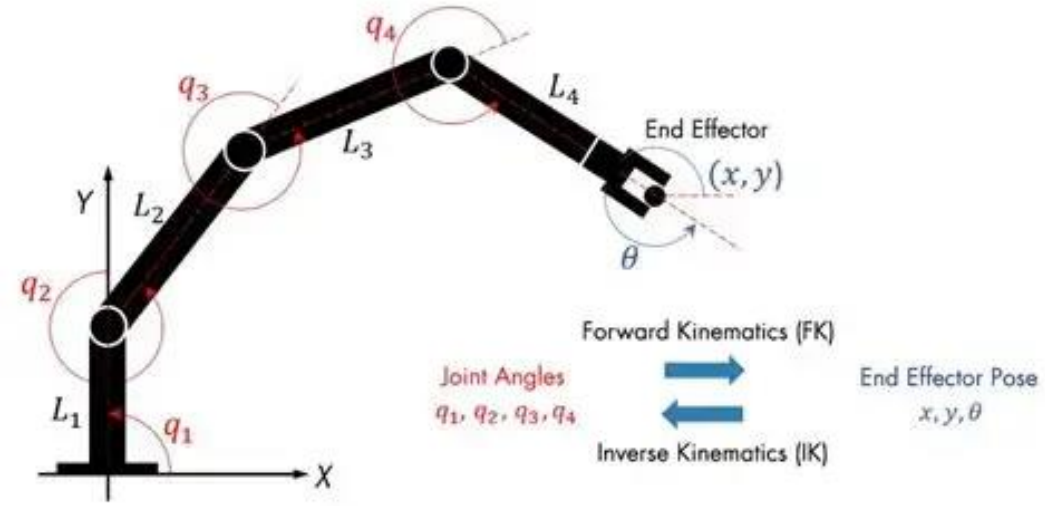





$z = z_1$
z_1
$P_1(r_1, \phi_1, z_1)$
$\mathbf{a}_z$
$\mathbf{a}_\phi$
$\mathbf{a}_r$
$r = r_1$ surface
r_1
y
ϕ_1
x
$\phi = \phi_1$ plane


z
$\hat{\mathbf{r}}$
$\hat{\boldsymbol{\varphi}}$
$\hat{\boldsymbol{\theta}}$
θ
r
O
φ
y
x

Mechanics





Kinematics

Dynamics

We analyze the motion and ignore the forces

Motion under the action of force

Motion in Straight Line (Simplest Motion)



Motion in Straight Line (Simplest Motion)



Motion in Straight Line

Scalars

- Distance $\rightarrow S$
- Speed $\rightarrow V$
  - Average Speed $\rightarrow V$
  - Instantaneous Speed $\rightarrow V$

Vectors

- Displacement $\rightarrow \vec{x}$
- Velocity $\rightarrow \vec{V}$
  - Average Velocity $\rightarrow ave\ \vec{V}$
  - Instantaneous Velocity $\rightarrow Ins\ \vec{V}$
- Acceleration $\rightarrow \vec{a}$
  - Average Acceleration $\rightarrow ave\ \vec{a}$
  - Instantaneous acceleration $\rightarrow Ins\ \vec{a}$

Motion in Straight Line

$Distance \rightarrow S$

$Displacement \rightarrow \vec{x}$

$S_R = x_2 - x_1 = 5\ m$

$\overrightarrow{\Delta X_R} = x_2 - x_1 = 10 - 5 = 5\ m$



$x_1 = 5\ m$

$x_2 = 10\ m$

Motion in Straight Line

$S_L = x_1 - x_1 = -5\ m$ $\qquad \overrightarrow{\Delta X_L} = x_1 - x_2 = 5 - 10 = -5\ m$

S → *Can never be negative*

$S_L = x_1 - x_2 = |-5|\ m$ $\qquad \overrightarrow{\Delta X_L} = 0$

$S_L = 5 + 5 = 10\ m$



$x_1 = 5\ m$ $\qquad x_2 = 10\ m$

Average Velocity

$$\overrightarrow{\Delta x} = x_2 - x_1 = 10 - 5 = 5\ m$$

$$\Delta t = t_2 - t_1 = 4 - 2 = 2\ s$$

$t_1 = 2\ s$ $t_2 = 4\ s$



$x_1 = 5\ m$ $x_2 = 10\ m$

Average Velocity

$$\overrightarrow{\Delta x} = \frac{\overrightarrow{\Delta x}}{\Delta t} = \frac{10 - 5}{4 - 2} = \frac{5}{2}\ \mathrm{m/s}$$

$t_1 = 2\ s$

$t_2 = 4\ s$



$x_1 = 5\ m$

$x_2 = 10\ m$

Average Velocity

$$\overrightarrow{\Delta v} = \frac{\overrightarrow{\Delta x}}{\Delta t} = \frac{10-5}{4-2} = \frac{5}{2}\ \mathrm{m/s}$$

$$\overrightarrow{\Delta v} = \frac{\overrightarrow{\Delta x}}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

$t_1 = 2\ s$

$t_2 = 4\ s$



$x_1 = 5\ m$

$x_2 = 10\ m$

Average Velocity vs Average Speed

$Velocity \rightarrow Vector$

$$\overrightarrow{\Delta v} = \frac{\overrightarrow{\Delta x}}{\Delta t}$$

$Speed \rightarrow Scalar$

$$\Delta v = \left| \frac{\Delta x}{\Delta t} \right|$$

$\overrightarrow{\Delta x}$ is zero in road trip

Δx can not be zero road trip

Average Velocity vs Average Speed

Sign of Velocity (v_x)	**Sign of Acceleration (a_x)**	**Effect on Motion**	**Example (Positive = Right)**
+ (Moving Right)	**+** (Acceleration to Right)	**Speeding Up**	Car moving right and pressing the gas.

Average Velocity vs Average Speed

Sign of Velocity (v_x)	**Sign of Acceleration (a_x)**	**Effect on Motion**	**Example (Positive = Right)**
+ (Moving Right)	**+** (Acceleration to Right)	**Speeding Up**	Car moving right and pressing the gas.
+ (Moving Right)	**–** (Acceleration to Left)	**Slowing Down**	Car moving right and pressing the brake.

Average Velocity vs Average Speed

Sign of Velocity (v_x)	Sign of Acceleration (a_x)	Effect on Motion	Example (Positive = Right)
+ (Moving Right)	**+** (Acceleration to Right)	**Speeding Up**	Car moving right and pressing the gas.
+ (Moving Right)	**–** (Acceleration to Left)	**Slowing Down**	Car moving right and pressing the brake.
– (Moving Left)	**–** (Acceleration to Left)	**Speeding Up**	Car moving left and pressing the gas (in the direction it's moving).

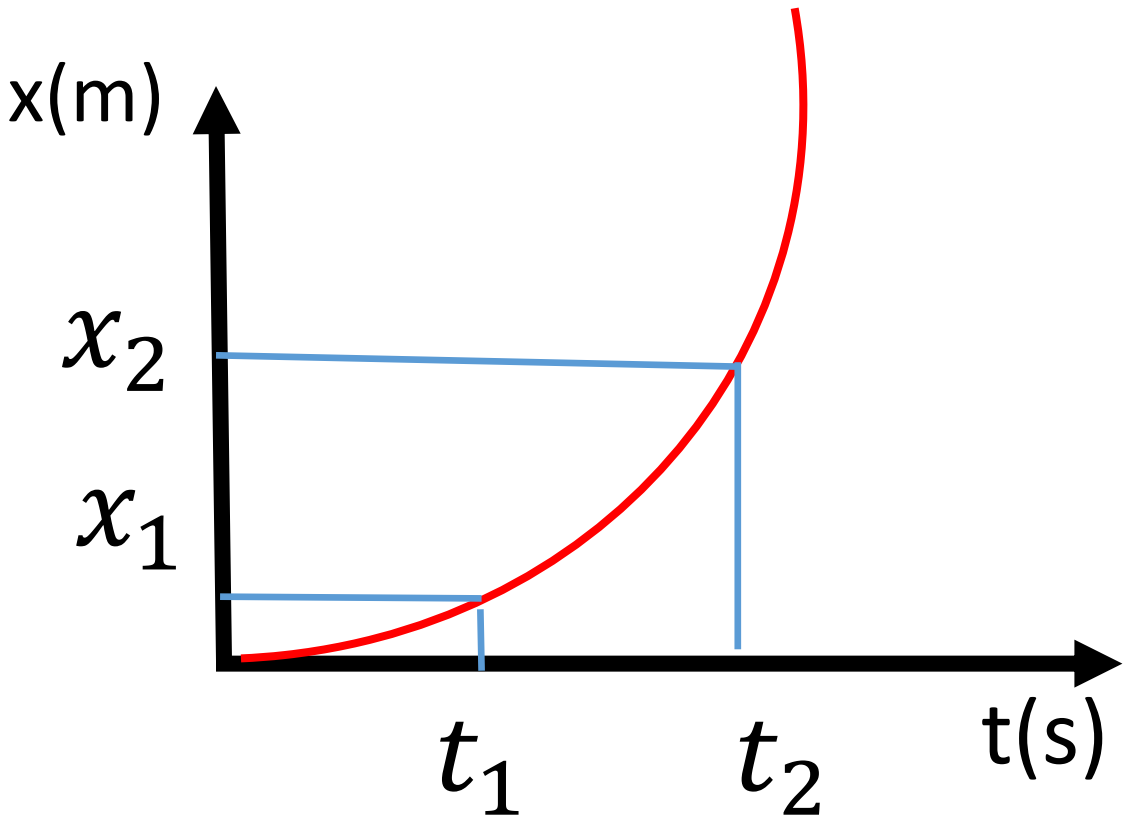
Average Velocity vs Average Speed

Sign of Velocity (v_x)	**Sign of Acceleration (a_x)**	**Effect on Motion**	**Example (Positive = Right)**
+ (Moving Right)	**+** (Acceleration to Right)	**Speeding Up**	Car moving right and pressing the gas.
+ (Moving Right)	**–** (Acceleration to Left)	**Slowing Down**	Car moving right and pressing the brake.
– (Moving Left)	**–** (Acceleration to Left)	**Speeding Up**	Car moving left and pressing the gas (in the direction it's moving).
– (Moving Left)	**+** (Acceleration to Right)	**Slowing Down**	Car moving left and pressing the brake (acceleration opposes motion).

If x-coordinate is:	**... x-velocity is:**
Positive & increasing (getting more positive)	Positive: Particle is moving in +x-direction
Positive & decreasing (getting less positive)	Negative: Particle is moving in –x-direction
Negative & increasing (getting less negative)	Positive: Particle is moving in +x-direction
Negative & decreasing (getting more negative)	Negative: Particle is moving in –x-direction

Average Velocity

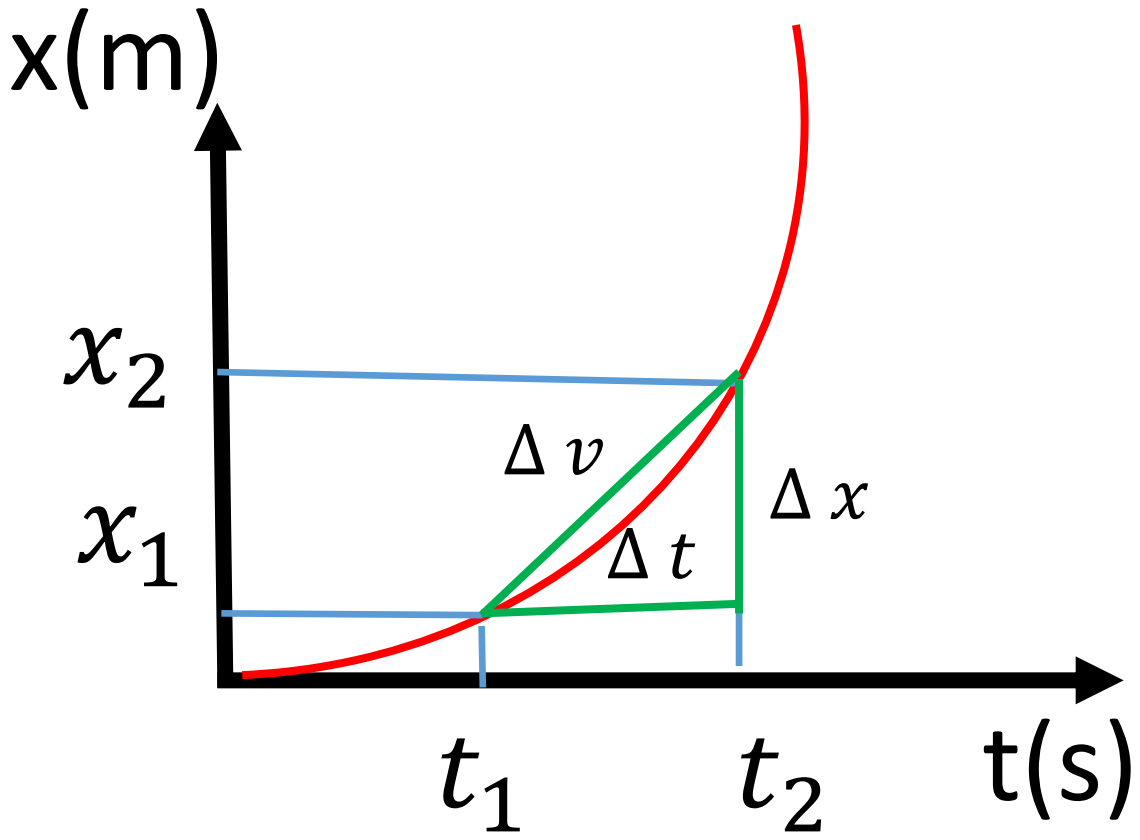
$x - t\ Graph \rightarrow$ Average Velocity



1. **Sign of v_x:** Determines direction of travel along the x-axis.
2. **From x-t Graph:** Sign of the slope = sign of the velocity.
3. **From v-t Graph:** Value on the y-axis = sign of the velocity.
4. **Speed Change:** Compare the sign of v_x and a_x.
   - Same Sign → Speeding Up
   - Opposite Signs → Slowing Down

Average Velocity

$x - t\ Graph \rightarrow$ Average Velocity



Slope$= \frac{Rise}{Run}$

$$\overrightarrow{\Delta v} = \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Instanteouns Velocity

$$\overrightarrow{\Delta v} = \lim_{\Delta t \to 0} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

$$\frac{dx}{dt} = \lim_{\Delta t \to 0} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

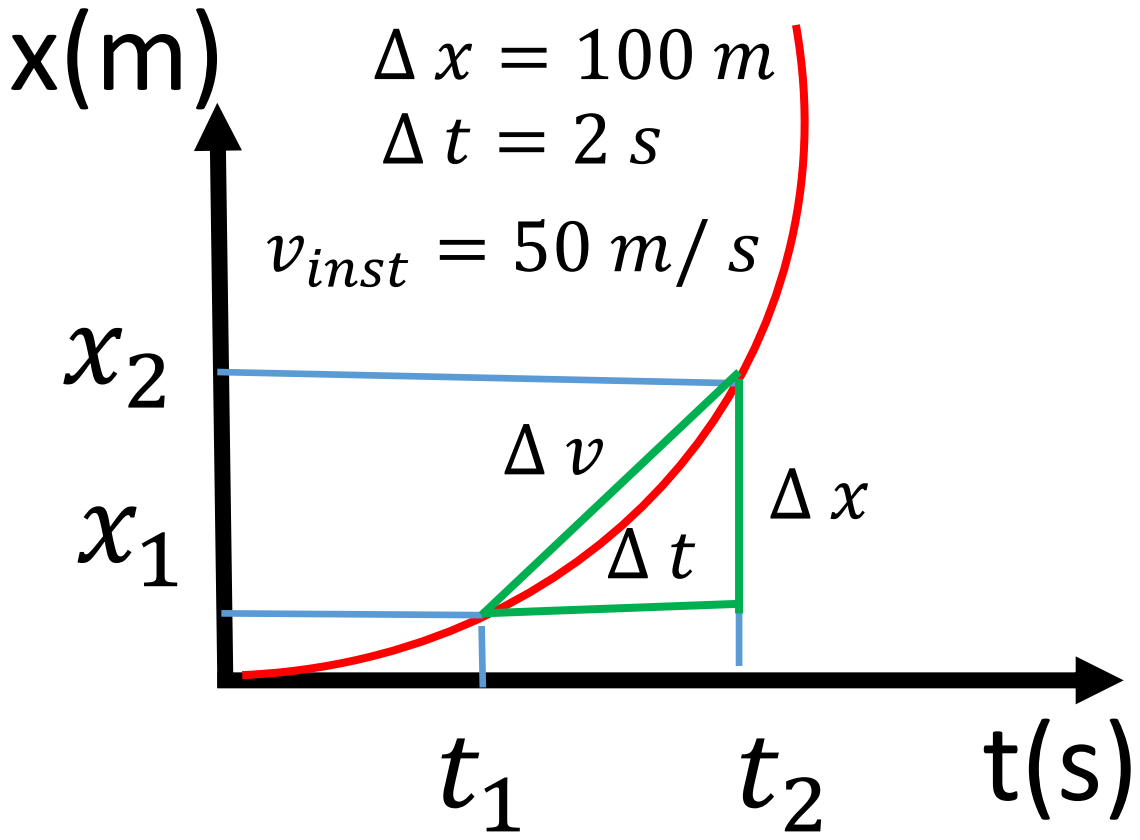
$$\overrightarrow{v_{inst}} = \lim_{\Delta t \to 0} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

$$\text{Slope} = \frac{Rise}{Run}$$

$$\overrightarrow{\Delta v} = \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Instanteouns Velocity

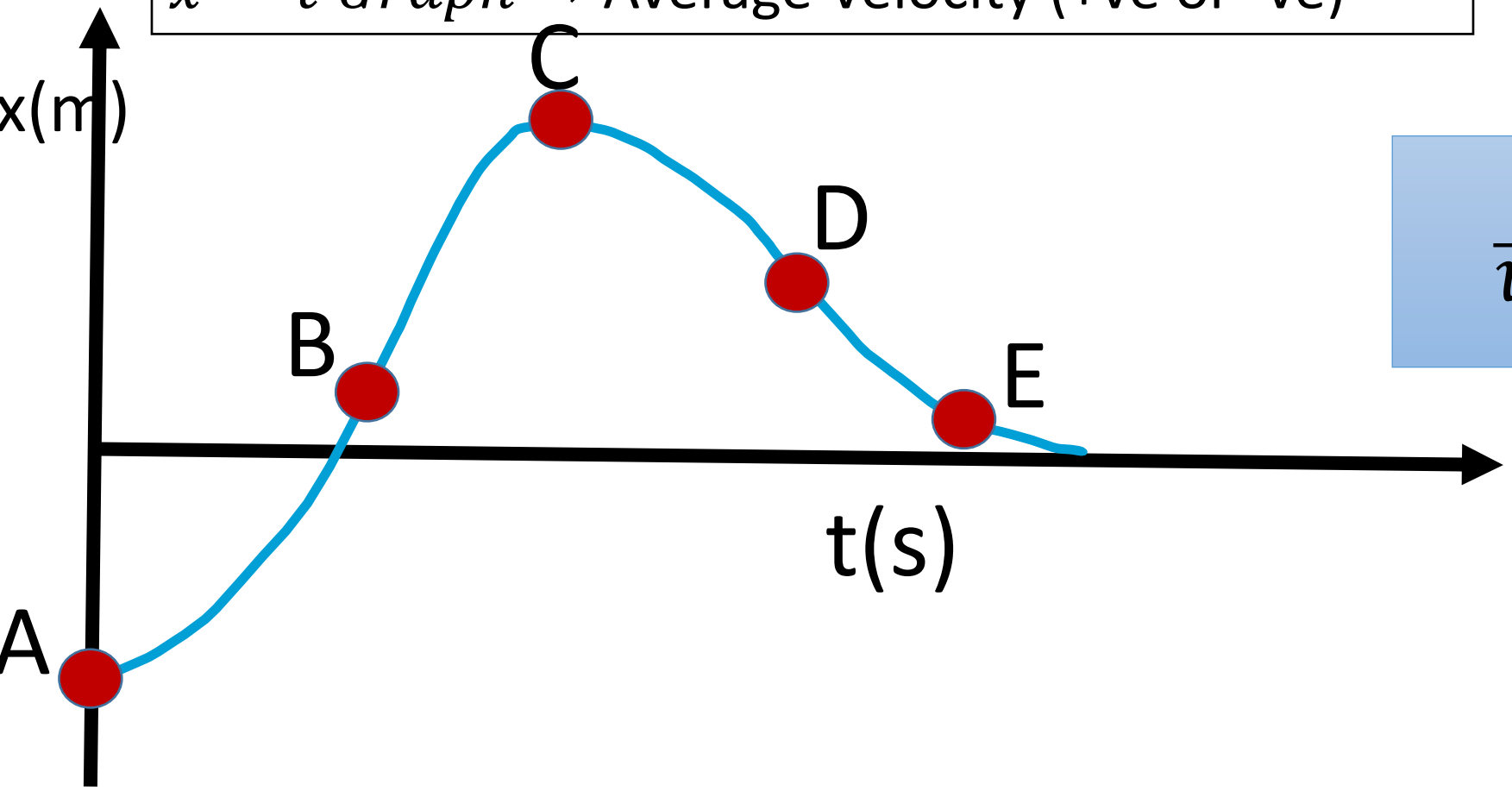
$x - t\ Graph \rightarrow$ Average Velocity



$$\overrightarrow{v_{inst}} = \lim_{\Delta t \to 0} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Instanteouns Velocity

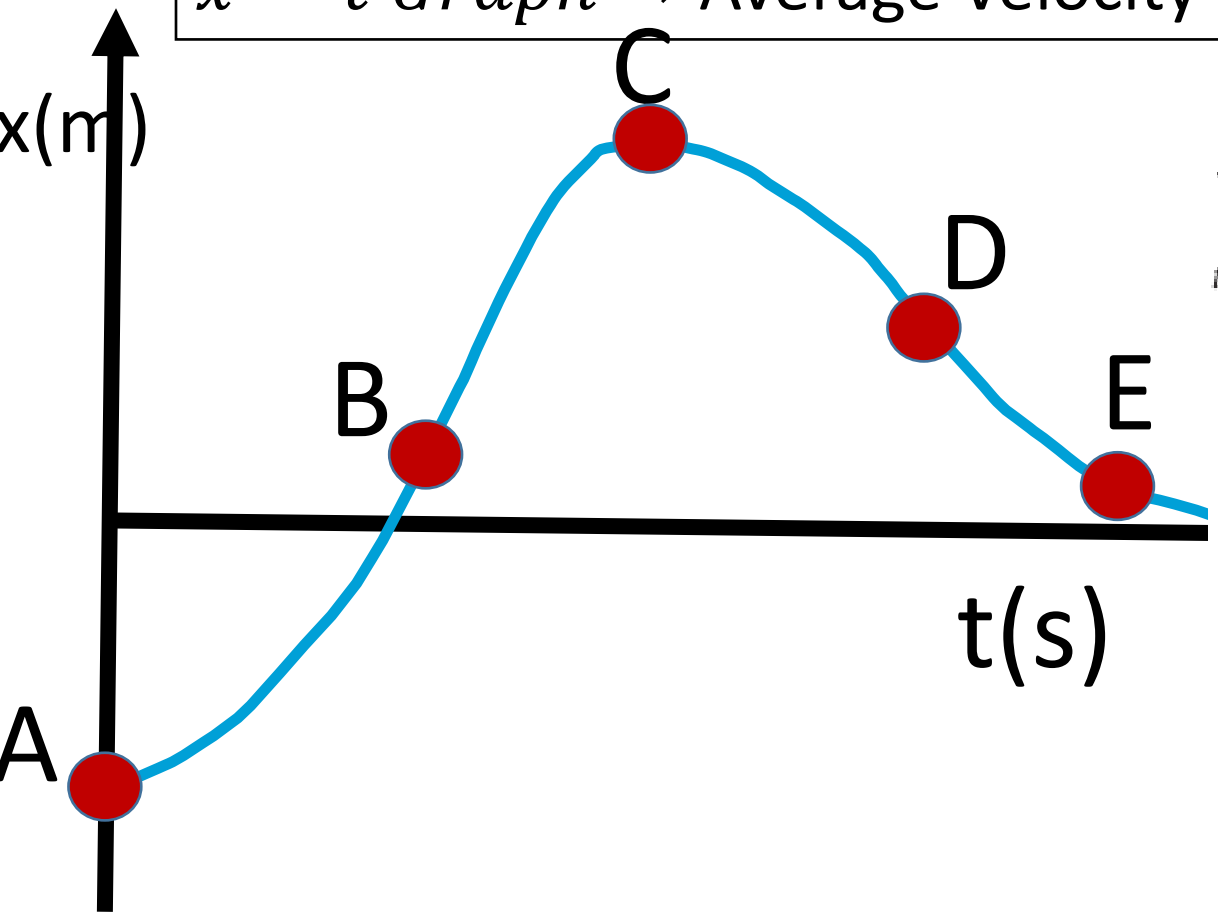
$x - t\ Graph \rightarrow$ Average Velocity (+ve or -ve)

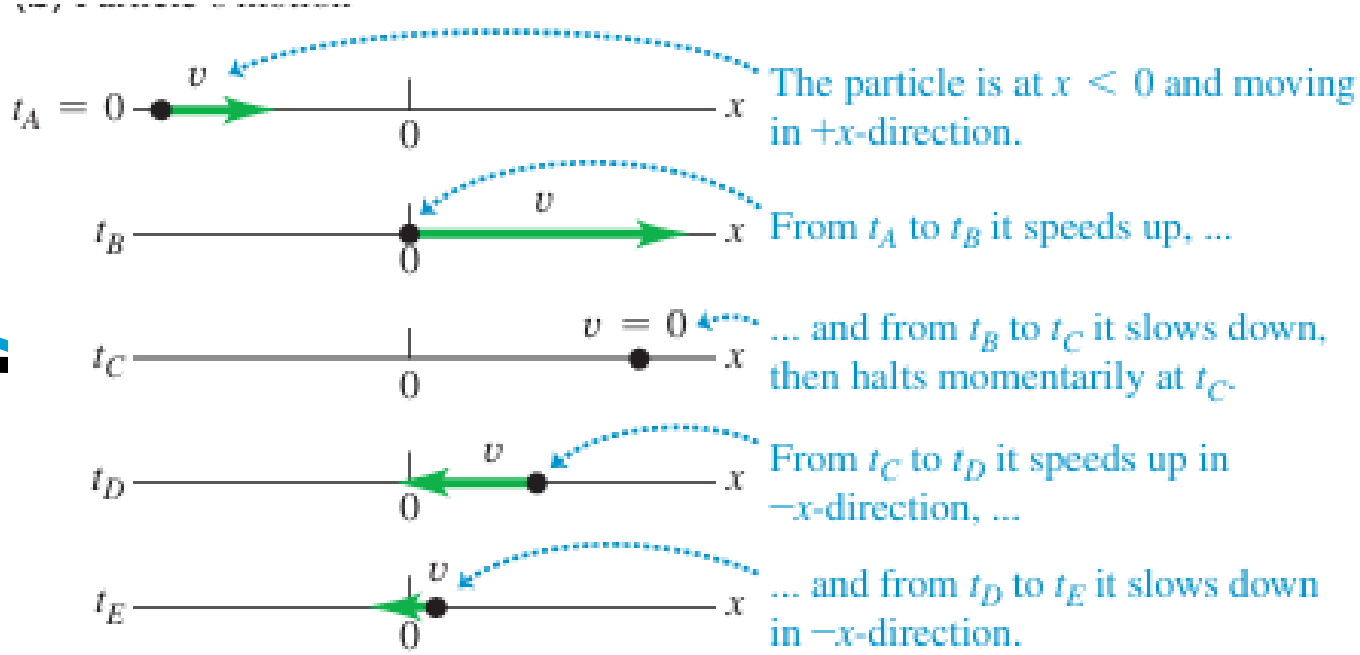


$$\overrightarrow{v_{inst}} = \lim_{\Delta t \to 0} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Instanteouns Velocity

$x - t\ Graph \rightarrow$ Average Velocity (+ve or -ve)

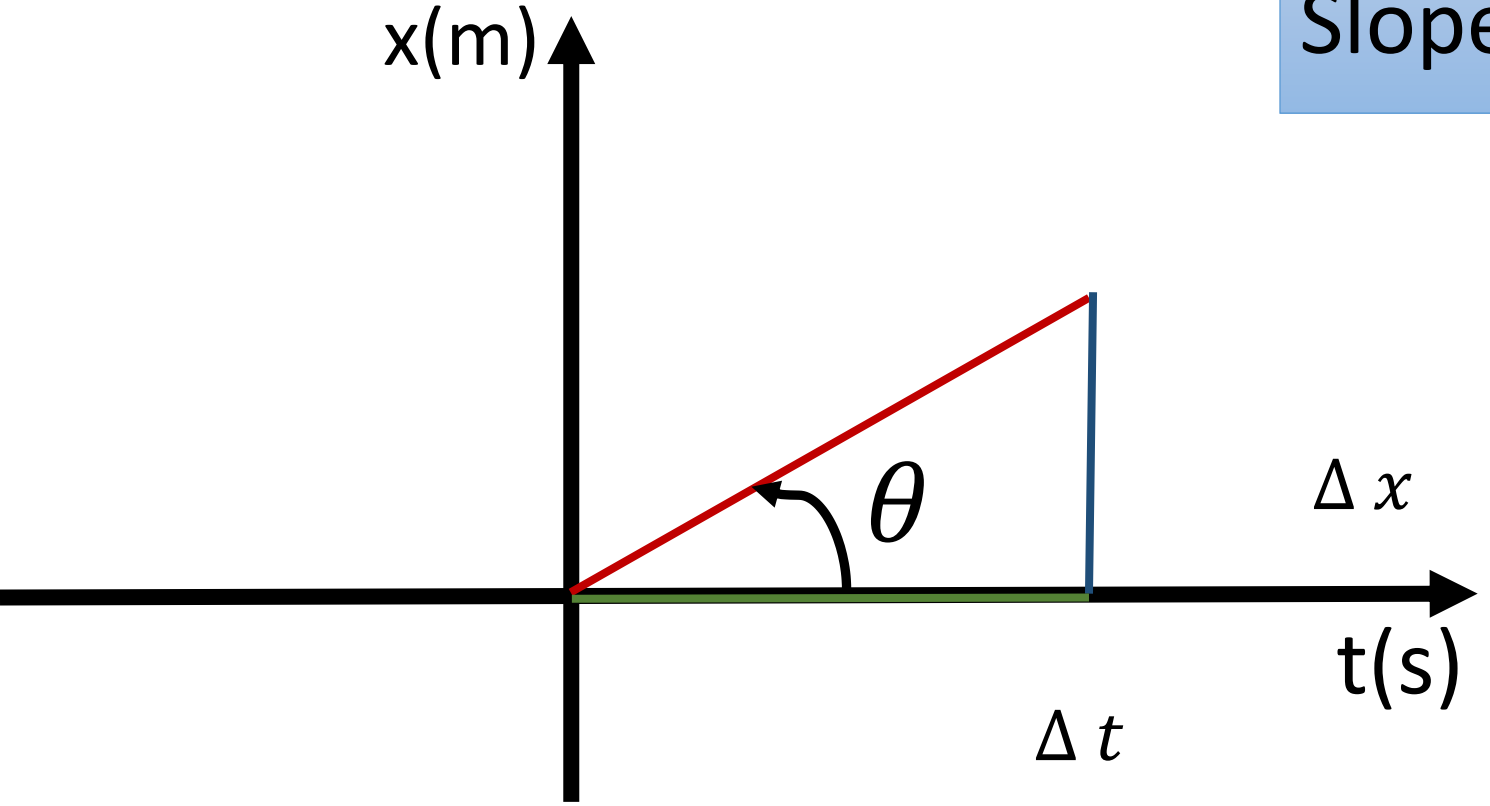




Average Velocity

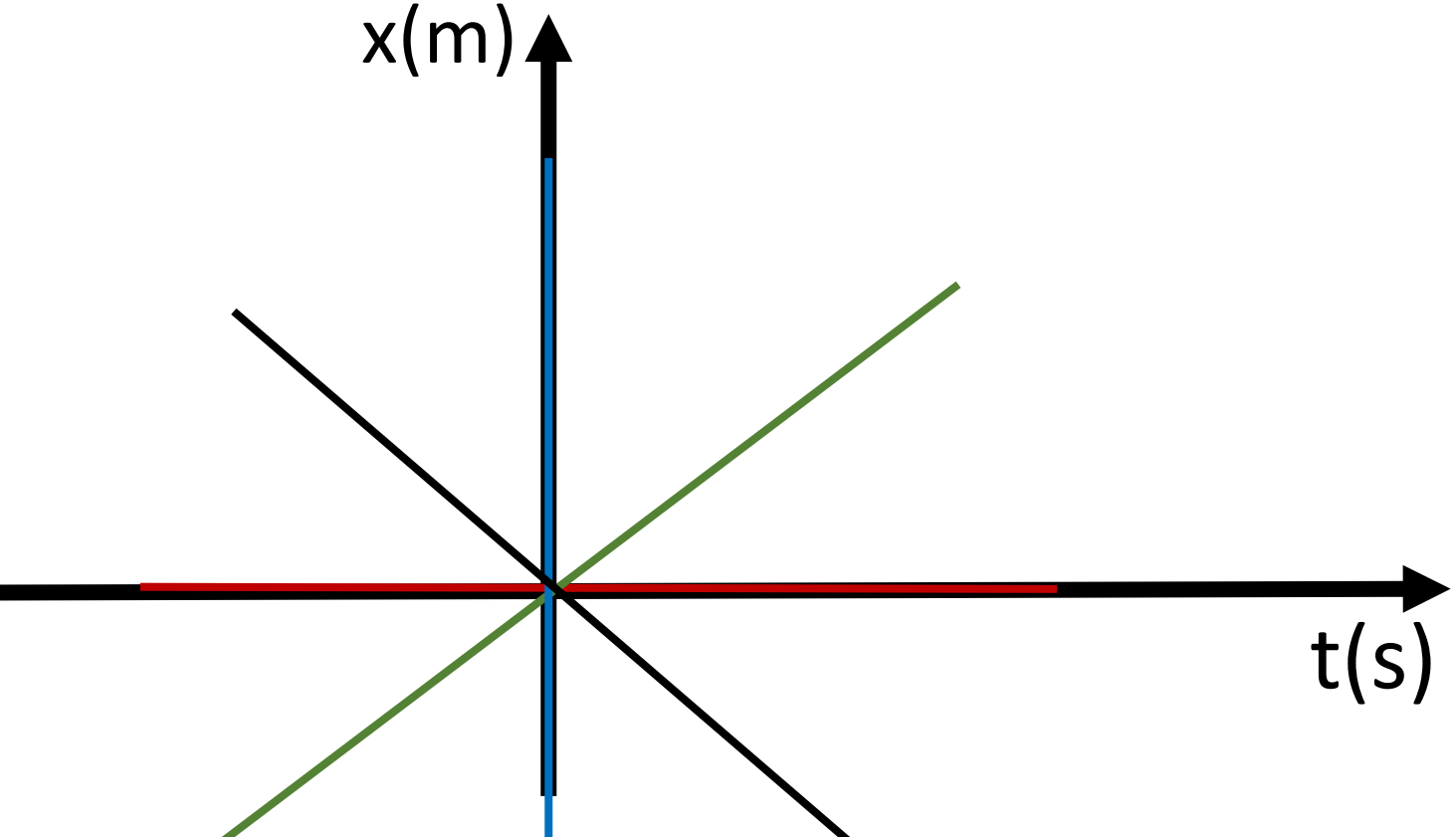
$x - t\ Graph \rightarrow$ Average Velocity

$$\text{Slope} = \frac{Rise}{Run} = \frac{Rise}{Run} = \tan(\theta)$$



Average Velocity

$x - t\ Graph \rightarrow$ Average Velocity



$\tan(0^o) = 0$

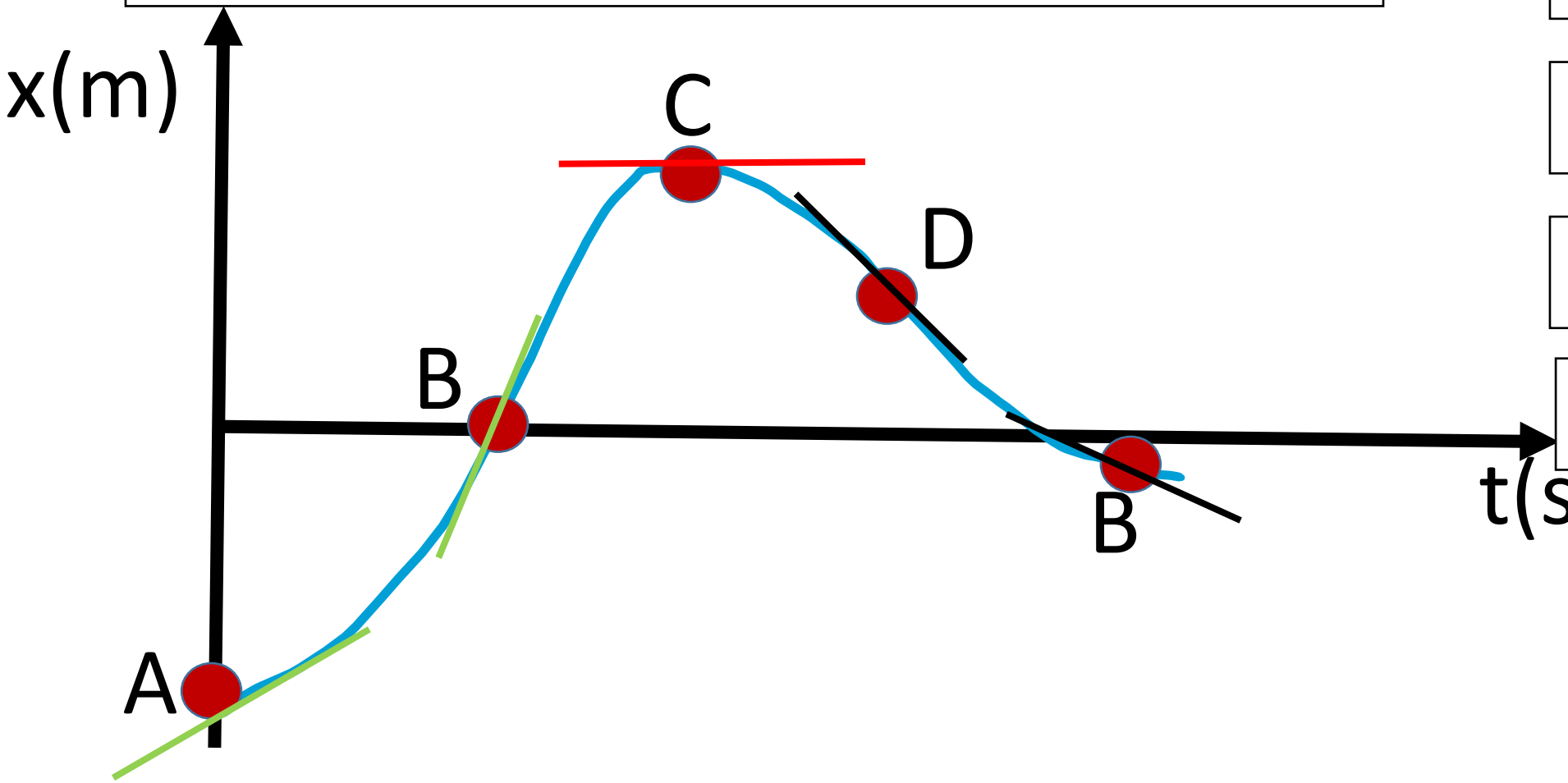
$\tan(\theta) > 0$

$\tan(90^o) = \infty$

$\tan(\theta) < 0$

Instantons Velocity Understanding

$x - t\ Graph \rightarrow$ Velocity (+ve or -ve)



$\tan(0^o) = 0$

$\tan(\theta) > 0$

$\tan(90^o) = \infty$

$\tan(\theta) < 0$

Four Kinematic Equations

Assumptions:

- Motion along a straight line (1D)
- Acceleration a is constant (uniform)
- At time $t = 0$: initial velocity $= u\ (or\ v_0)$, initial position $= x_0$
- At time t: final velocity $= v$, final position $= x$

Acceleration (constant):

$$a = (v - u) / t$$

Average velocity:

$$v_avg = (u + v) / 2$$

Four Kinematic Equations (**First Equation of Motion**)

Acceleration (constant):

$$a = (v - u) / t$$

$$at = v - u$$

$$v = u + at$$

Four Kinematic Equations (**Second Equation of Motion**)

Average velocity:

$$v_avg = (u + v) / 2$$

$$s = v_avg \times t$$

$$s = [(u + v) / 2] \times t$$

$$s = [(u + (u + at)) / 2] \times t$$

$$s = [(2u + at) / 2] \times t$$

$$s = ut + \frac{1}{2}at^2$$

Four Kinematic Equations (**Third Equation of Motion**)

$$v = u + at \rightarrow$$
$$\mathbf{t = (v - u) / a}$$

$$s = ut + \tfrac{1}{2}at^2$$

$$s = u[(v - u)/a] + \tfrac{1}{2}a[(v - u)/a]^2$$

$$s = (uv - u^2)/a + (v - u)^2/(2a)$$

$$s = [2(uv - u^2) + (v - u)^2] / (2a)$$

$$2as = v^2 - u^2$$

Four Kinematic Equations (**Fourth Equation of Motion**)

$$v_avg = (u + v) / 2$$

$$v = u + at$$

$$v_avg = [(u + at) + v] / 2 = (2u + at) / 2$$

$$s = (2ut + at^2) / 2$$

$$s = ut + \tfrac{1}{2}at^2$$

Example

A GIK student leaves his hostel and walks along the sidewalk toward campus. After 5 min it starts to rain, and he returns hostel. His distance from her hostel as a function of time is shown in **Fig.**

At which of the labeled points is her velocity

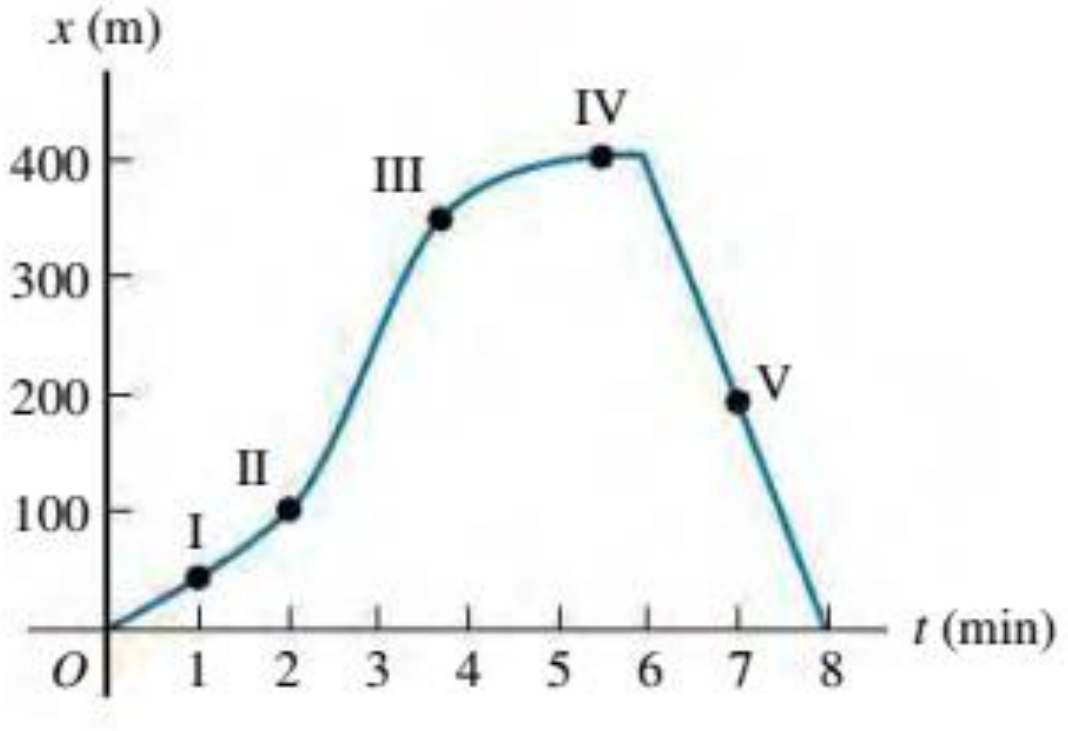
(a) zero? IV

(b) constant and positive? I

(c) constant and negative? V

(d) increasing in magnitude? II

(e) decreasing in magnitude? III



Example

A cheetah is crouched 20 m to the east of a vehicle (Fig). At time t = 0 the cheetah begins to run due east toward an antelope that is 50 m to the east of the vehicle. During the first 2.0 s of the chase, the cheetah's x-coordinate varies with time according to the equation

$$x = 20\ m + (5\ m/s^2)t^2$$

(a) Find the cheetah's displacement be tween t_1 = 1.0 s and t_2 = 2.0 s.

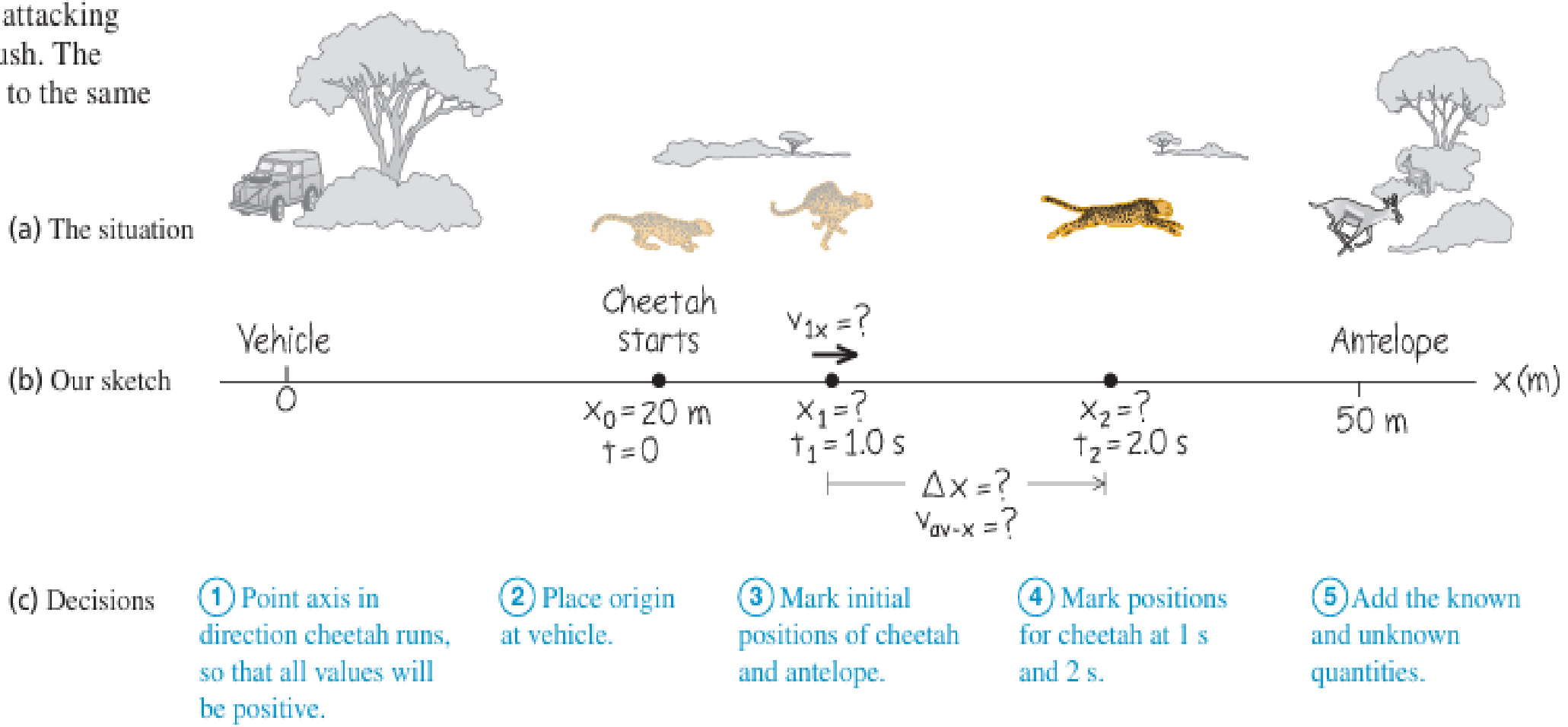
(b) Find its average velocity during that interval.

(c) Find its instantaneous velocity at t_1 = 1.0 s by taking Δt = 0.1 s, then 0.01 s, then 0.001 s.

(d) Derive an expression for the cheetah's instantaneous velocity as a function of time, and use it to find v_x at t = 1.0 s and t = 2.0 s.

Example

Figure 2.6 A cheetah attacking an antelope from ambush. The animals are not drawn to the same scale as the axis.



Example

Find the cheetah's displacement be tween t_1 = 1.0 s and t_2 = 2.0

(a)

$$x = 20\ m + (5\ m/s^2)t^2$$

$$x_1 = 20\ m + (5\ m/s^2)(1)^2 = 25\ \text{m}$$

$$x_2 = 20\ m + (5\ m/s^2)(2)^2 = 40\ \text{m}$$

$$\Delta x = 40 - 25 = 15\ m$$

$$\Delta t = 2 - 1 = 1\ s$$

Find its average velocity during that interval.

(b)

$$V_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{15}{1} = 15\ m/s$$

Example

(c) Find its instantaneous velocity at t_1 = 1.0 s by taking Δt = 0.1 s, then 0.01 s, then 0.001 s.

$\Delta t = 0.1\ s$ $t_1 = 1\ s$ $t_2 = 1.1\ s$

$x_1 = 20\ m + (5\ m/s^2)(1)^2 = 25$ m

$x_2 = 20\ m + (5\ m/s^2)(1.1)^2 = 26.05$ m

$V_{av} = \frac{26.05-25}{1.1-1} = 10.5\ m/s$

(d) Derive an expression for the cheetah's instantaneous velocity as a function of time, and use it to find v_x at t = 1.0 s and t = 2.0 s.

$V_x = \frac{dx}{dt} = \frac{d(20\ m+(5\ m/s^2)t^2\ )}{dt} = \left(10.5\frac{m}{s^2}\right)t$

$V(t=1) = (10.5 * 1) = 10.5\ m/s$

$V(t=2) = (10.5 * 2) = 21\ m/s$

Velocity vs Acceleration

Average Velocity

$$\overrightarrow{v_{avg}} = \frac{\overrightarrow{\Delta x}}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Average Acceleration

$$\overrightarrow{a_{avg}} = \frac{\overrightarrow{\Delta v}}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$$

Instantaneous Velocity

$$\overrightarrow{v_{inst}} = \frac{dx}{dt} = \lim_{\Delta t \to \infty} \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Instantaneous Acceleration

$$\overrightarrow{a_{inst}} = \frac{dv}{dt} = \lim_{\Delta t \to \infty} \frac{\overrightarrow{\Delta v}}{\Delta t}$$

Important Points

$$if\,\Delta x > 0\ object\ is\ moving\ toward\ the + ve\ x - axis$$

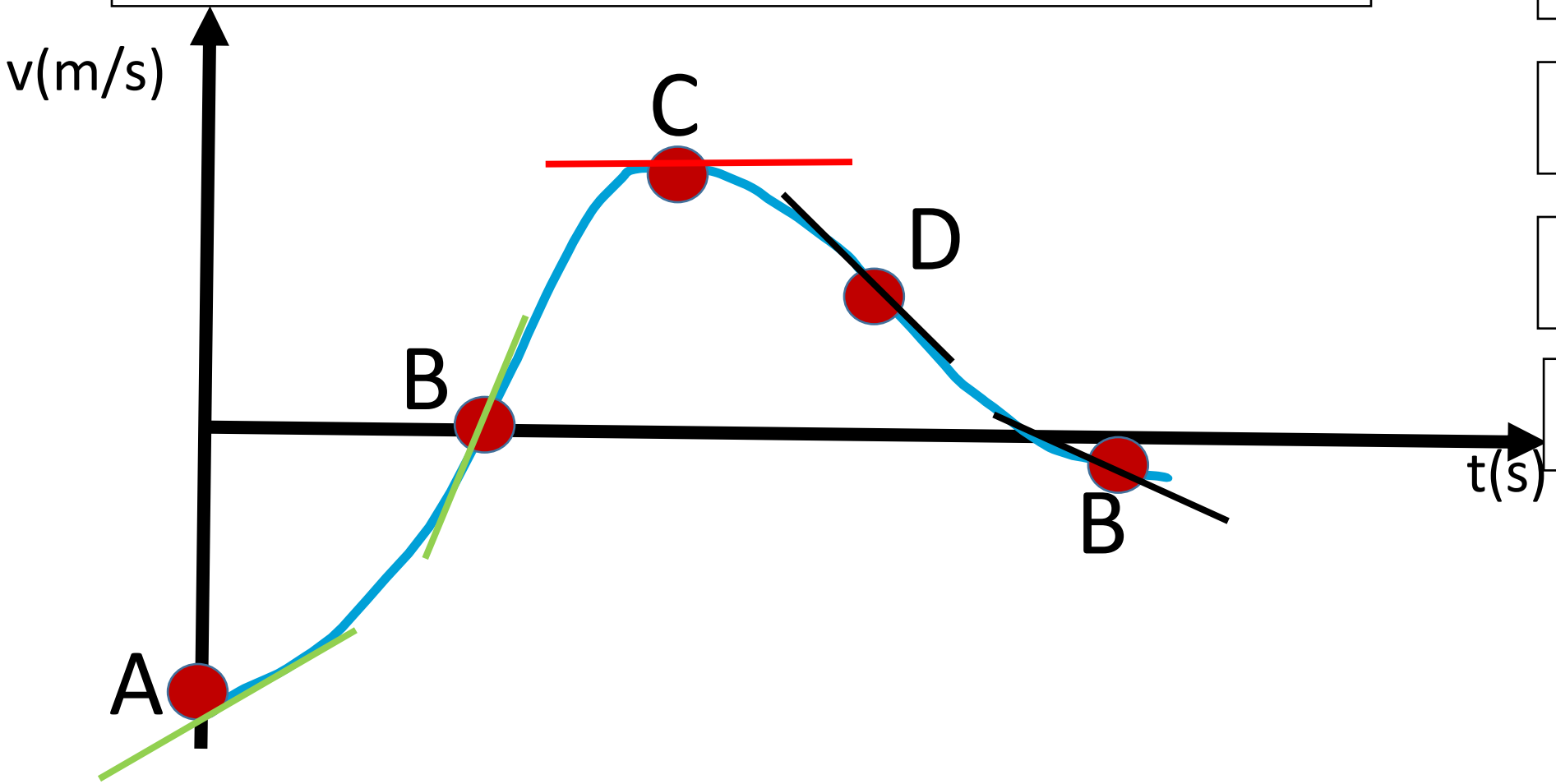
$$if\,\vec{v} > 0\ object\ is\ moving\ toward\ the + ve\ x - axis$$

$$speed \neq \left|\frac{\overrightarrow{\Delta x}}{\Delta t}\right|$$

In one-domentional motion, we assume that motion only along x-axis **or** y-axis **or** z-axis

Instanteouns Acceleration Understanding

$v - t\ Graph \rightarrow$ Acceleration (+ve or -ve)



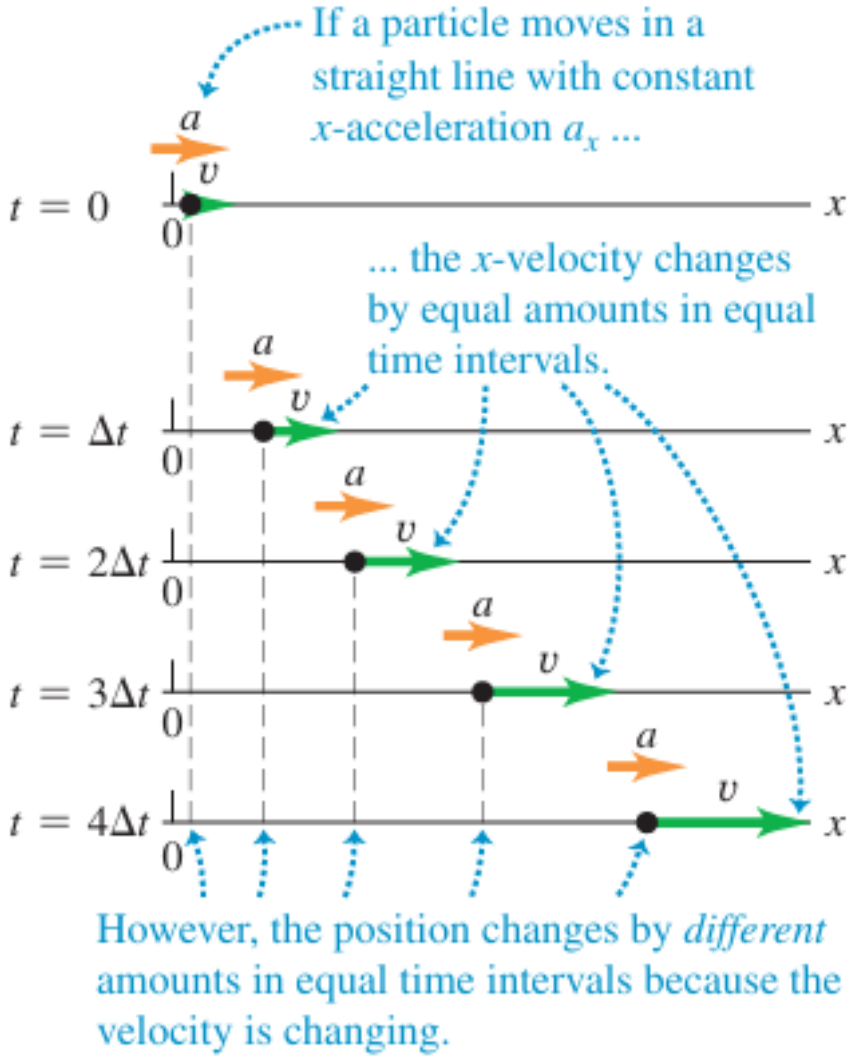
$\tan(0^o) = 0$

$\tan(\theta) > 0$

$\tan(90^o) = \infty$

$\tan(\theta) < 0$

Motion with constant acceleration

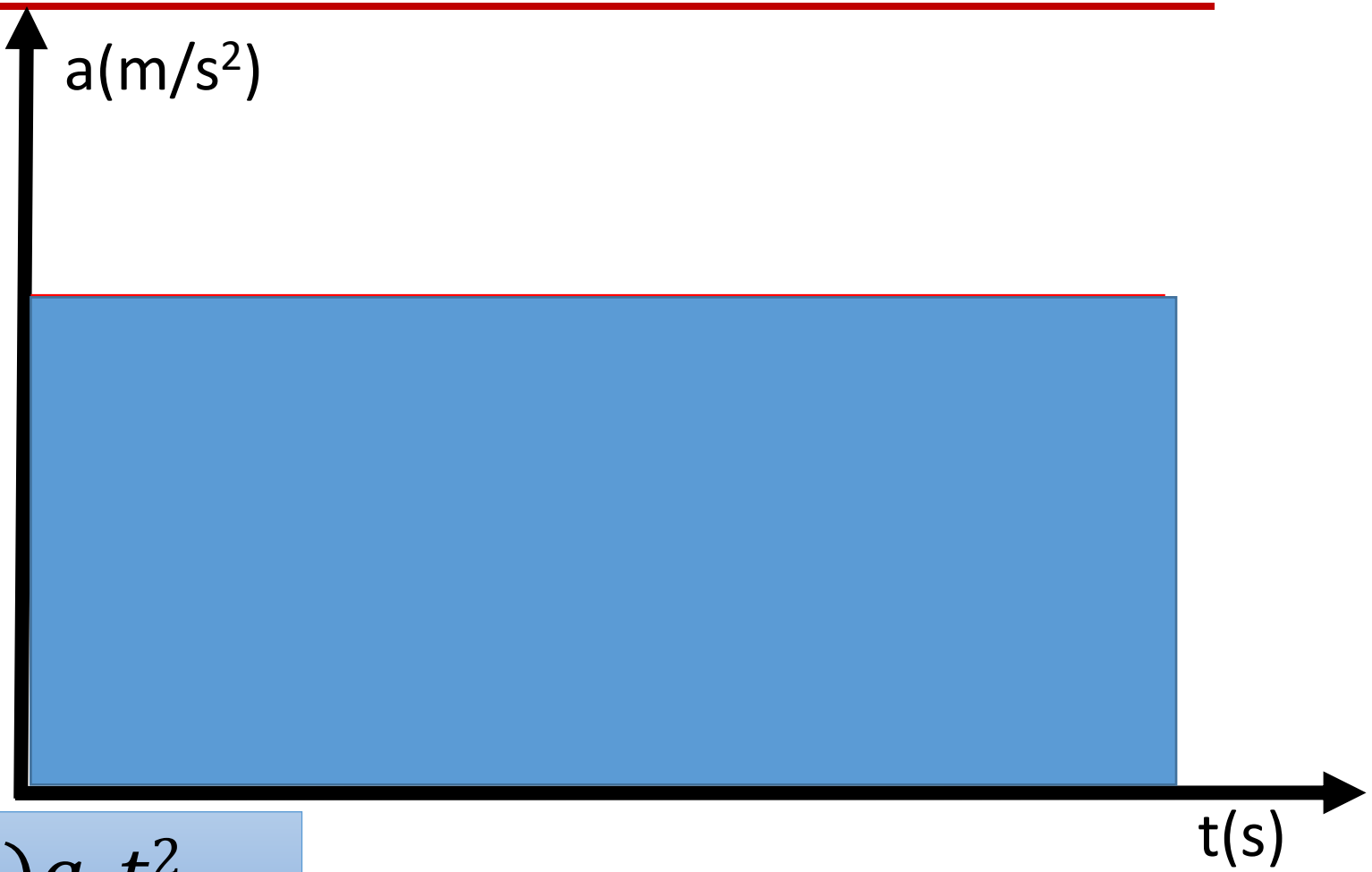


Motion with constant acceleration

$$\vec{a} = \frac{\overrightarrow{\Delta v}}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$$

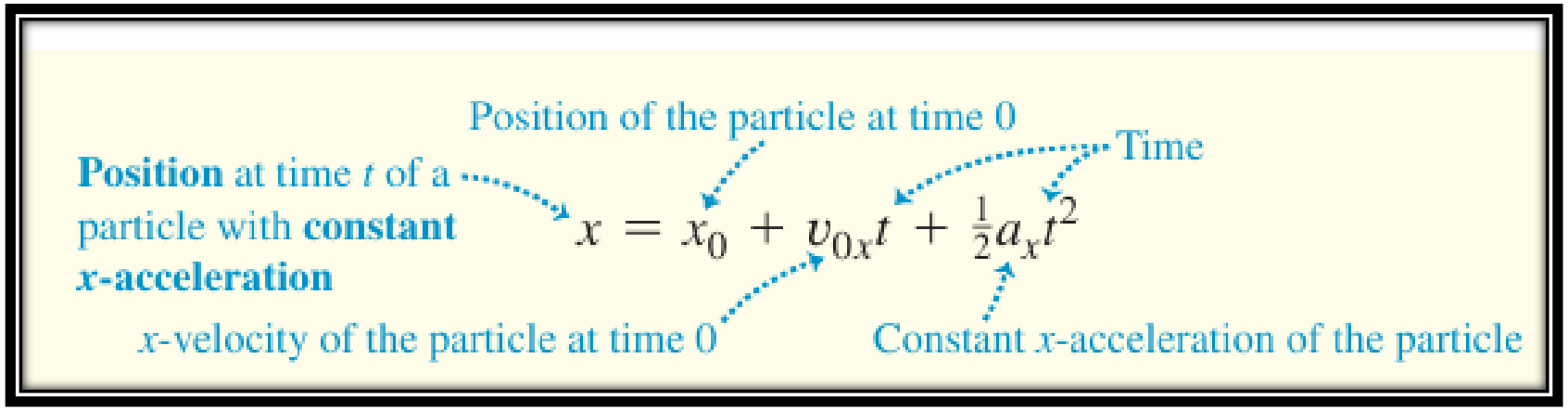
$$v_x = v_0 + at$$

$$x = x_0 + v_{0x}t + (1/2)a_x t^2$$



Motion with constant acceleration

$$v_x = v_0 + at$$



$$x = x_0 + v_{0x}t + (1/2)a_x t^2$$

Equations of Motion with Constant Acceleration

$$v_x = v_{0x} + at$$

$$x = x_0 + v_{0x}t +$$

$$v_x^2 = v_{0x}^2 + 2a$$

$$x - x_0 = (1/2)(v_x$$

TABLE 2.5 Equations of Motion with Constant Acceleration

Equation		Includes Quantities			
$v_x = v_{0x} + a_x t$	(2.8)	t		v_x	a_x
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	(2.12)	t	x		a_x
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	(2.13)		x	v_x	a_x
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	(2.14)	t	x	v_x	

Free falling Object

$$v_x = v_{0x} + at$$

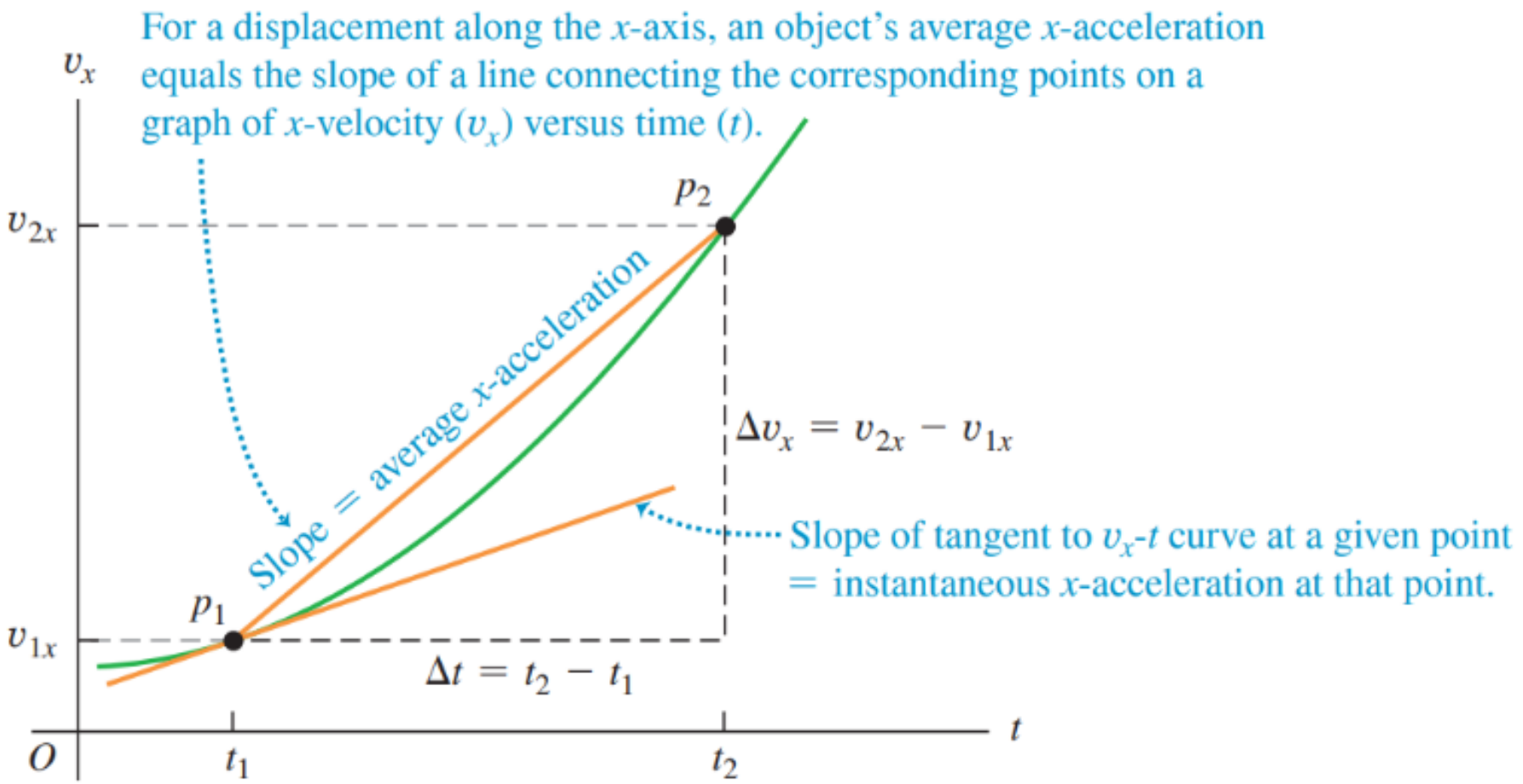
Acceleration: $a_y = -g$(constant downward acceleration)

$$x = x_0 + v_{0x}t + (1/2)a_x t^2$$

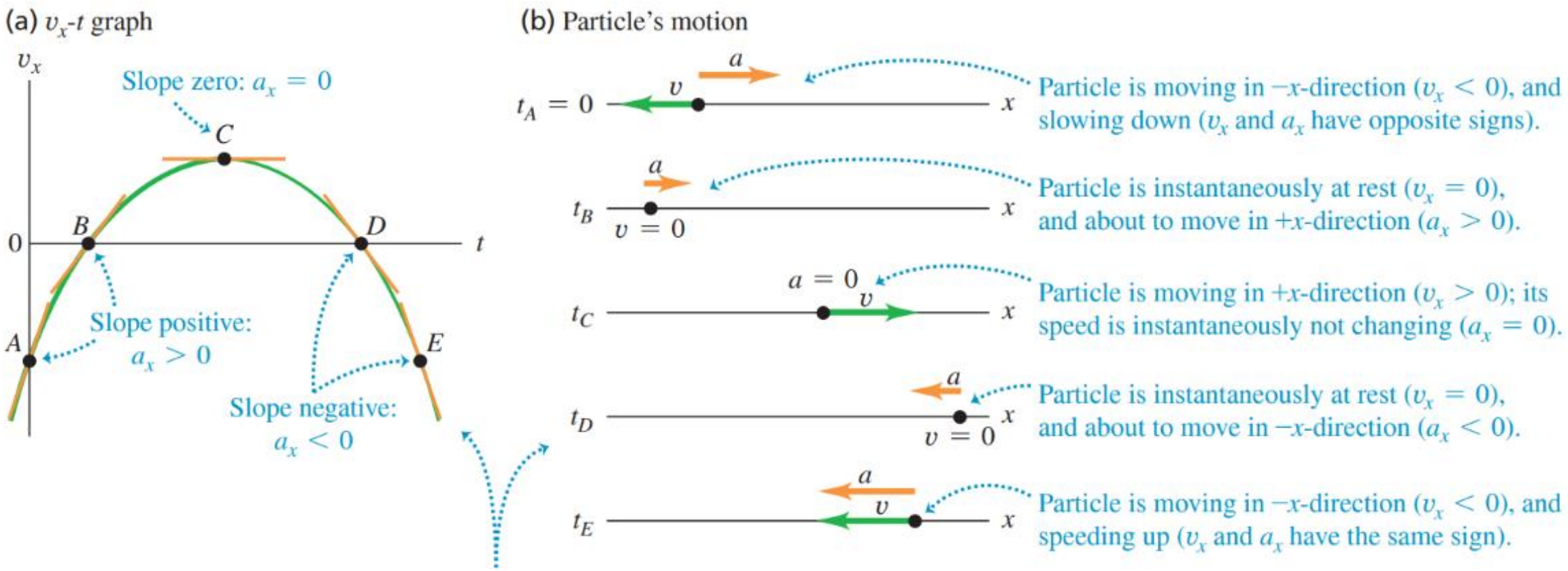
$$v_x{}^2 = v_{0x}{}^2 + 2a_x(x - x_0)$$

$$x - x_0 = (1/2)(v_x + v_{0x})t$$

Finding Acceleration on a v_x-t Graph or an x-t Graph



Finding Acceleration on a v_x-t Graph or an x-t Graph



Equations of Motion with Constant Acceleration

Equation	Missing Variable	Use Case
1. $v_x = v_{0x} + a_x t$	**Displacement** ($x - x_0$)	Finding velocity after a given time, without needing
2. $x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$	**Final Velocity** (v_x)	Finding position after a given time.
3. $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	**Time** (t)	Relating velocity to position, when time is unknown
4. $x = x_0 + \frac{1}{2}(v_{0x} + v_x)t$	**Acceleration** (a_x)	Finding displacement using the average velocity.

Equations of Motion with Constant Acceleration

Projectile motion is a direct application of the 1D equations to two dimensions (x and y) independently.

The acceleration is constant: $a_x = 0$ and $a_y = -g$.

Position as a function of time:

- $x = x_0 + (v_0 \cos\theta)t$
- $y = y_0 + (v_0 \sin\theta)t - \frac{1}{2}gt^2$

Velocity as a function of time:

- $v_x = v_0 \cos\theta$ (constant)
- $v_y = v_0 \sin\theta - gt$

Projectile motion = Horizontal motion (uniform velocity) + Vertical motion (uniform acceleration)

Equations of Motion with Constant Acceleration

Projectile motion = Horizontal motion (uniform velocity) + Vertical motion (uniform acceleration)

These two motions:

- Happen simultaneously
- Do not affect each other
- Can be solved separately
- Are combined only at the end for the final answer

Gravity Acts Only Vertically

Gravity pulls the projectile downward only. It has:

No horizontal component ($g_x = 0$)

Only vertical component ($g_y = -9.8\ m/s^2$)

The Two Possible Sign Conventions

You have two choices for your 1D axis:

Convention A: Upward is Positive (+)

- g is negative: $a = -g = -9.8\ m/s^2$
- Downward motion → negative velocity
- Objects thrown up → positive initial velocity

Convention B: Downward is Positive (+)

- g is positive: $a = +g = +9.8\ m/s^2$
- Downward motion → positive velocity
- Objects thrown up → negative initial velocity

Both conventions give the same physical answers.

Example

A turtle crawls along a straight line, which we'll call the x-axis with the positive direction to the right. The equation for the turtle's position as a function of time is

$$x = 50\ cm + (2\ cm/s)t - (0.0625\ cm/s^2)t^2$$

(a) Find the turtle's initial velocity, initial position, and initial acceleration.
(b) At what time "t" is the velocity of the turtle zero?
(c) How long after starting does it take the turtle to return to its starting point?
(d) At what times "t" is the turtle a distance of 10.0 cm from its starting point? What is the velocity (magnitude and direction) of the turtle at each of those times?
(e) Sketch graphs of **x versus t**, **v_x versus t**, and **a_x versus t**, for the time interval t = 0 to t = 40 s.

Example

$$x = 50\ cm + (2\ cm/s)t - (0.0625\ cm/s^2)t^2$$

$$v_x = 2\ cm/s - (0.125\ cm/s^2)t$$

$$a_x = -(0.125\ m/s^2)$$

Find the turtle's initial velocity, initial position, and initial acceleration.

(a) $x = 50\ cm$ $v_x = 2\ cm/s$ $a_x = -0.125\ cm/s^2$

At what time "t" is the velocity of the turtle zero?

(b) $v_x = 0 = 2\ cm/s - (0.125\ cm/s^2)t$

$t = 16\ s$

Example

(c) How long after starting does it take the turtle to return to its starting point?

$$x = 50\ cm + (2\ cm/s)t - (0.0625\ cm/s^2)t^2$$

$$x = 50\ cm\ \ \&$$

$$t = 0\ s\ \&\ t = 32\ s$$

(d) At what times "t" is the turtle a distance of 10.0 cm from its starting point? What is the velocity (magnitude and direction) of the turtle at each of those times?

It is at 10 cm when x = 60 cm or x = 40 cm

x = 60 cm -> t = 6.2 s & t = 25.8 s

v_x = +1.23 cm/s &
v_x = -1.23 cm/s

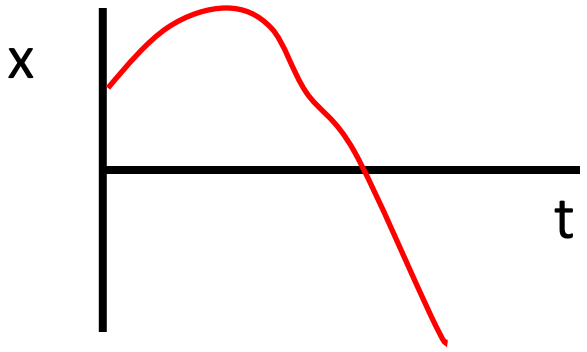
Example

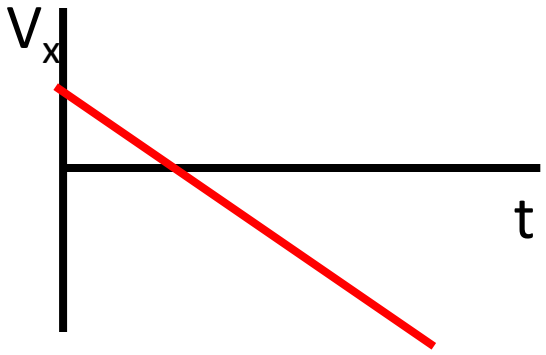
(e) Sketch graphs of **x versus t**, **v_x versus t**, and **a_x versus t**, for the time interval t = 0 to t = 40 s.

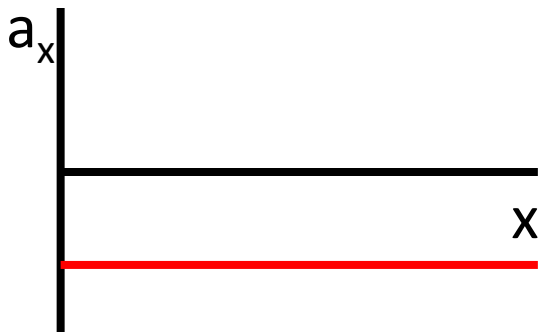
$$x = 50\ cm + (2\ cm/s)t - (0.0625\ cm/s^2)t^2$$

$$v_x = 2\ cm/s - (0.125\ cm/s^2)t$$

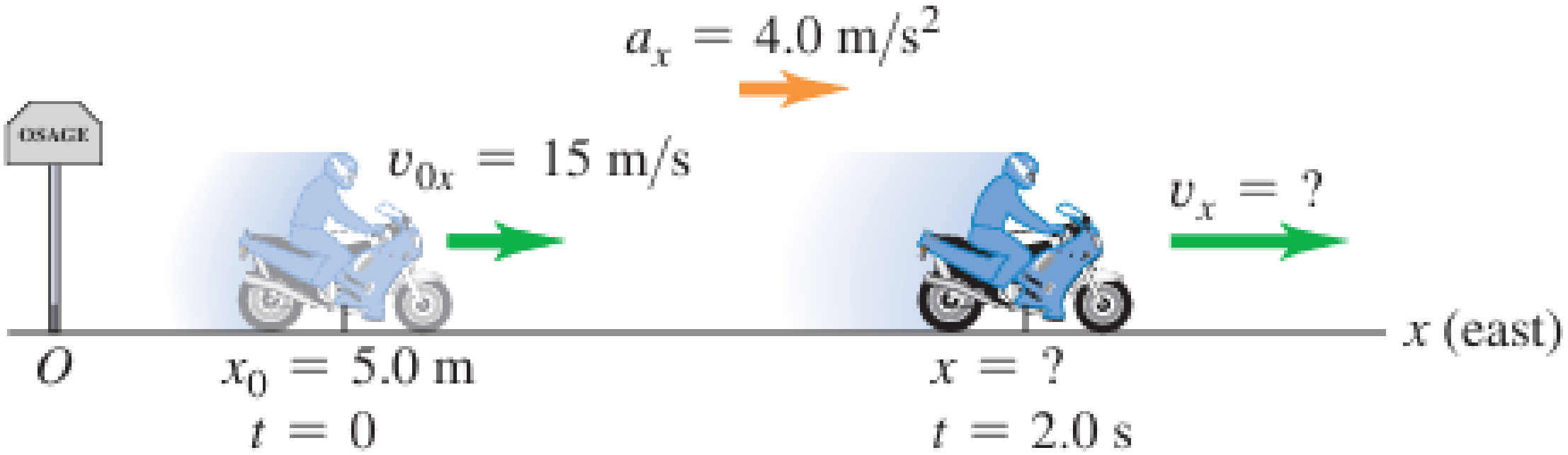
$$a_x = -(0.125\ m/s^2)$$







A motorcyclist traveling with constant acceleration.



A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits as shown in figure. At time t = 0 he is 5.0 m east of the city-limits signpost while he moves east at 15 m/s.

(a) Find his position and velocity at t = 2.0 s.

(b) Where is he when his speed is 25 m/s?

We take the signpost as the origin of coordinates (x = 0) and choose the positive x-axis to point. The known variables are the initial position and velocity, $x_0 = 5.0$ m and $v_{0x} = 15$ m/s, and the acceleration, $a_x = 4.0$ m/s^2. The unknown target variables in part (a) are the values of the position x and the x-velocity v_x at t = 2.0 s; the target variable in part (b) is the value of x when $v_x = 25$ m/s.

$$x = x_0 + v_{0x}t + (1/2)a_x t^2$$

$$x = 5 + (15)(2) + (1/2)(4)(2)^2 = 43\ m$$

$$v_x = v_{0x} + at$$

$$v_x = 15 + (4)(2) = 23\text{m/s}$$

We take the signpost as the origin of coordinates (x = 0) and choose the positive x-axis to point. The known variables are the initial position and velocity, x_0 = 5.0 m and v_{0x} = 15 m/s, and the acceleration, a_x = 4.0 m/s^2. The unknown target variables in part (a) are the values of the position x and the x-velocity v_x at t = 2.0 s; the target variable in part (b) is the value of x when v_x = 25 m/s.

Value of x when v_x = 25 m/s, but we don't know the time when the motorcycle has this velocity.

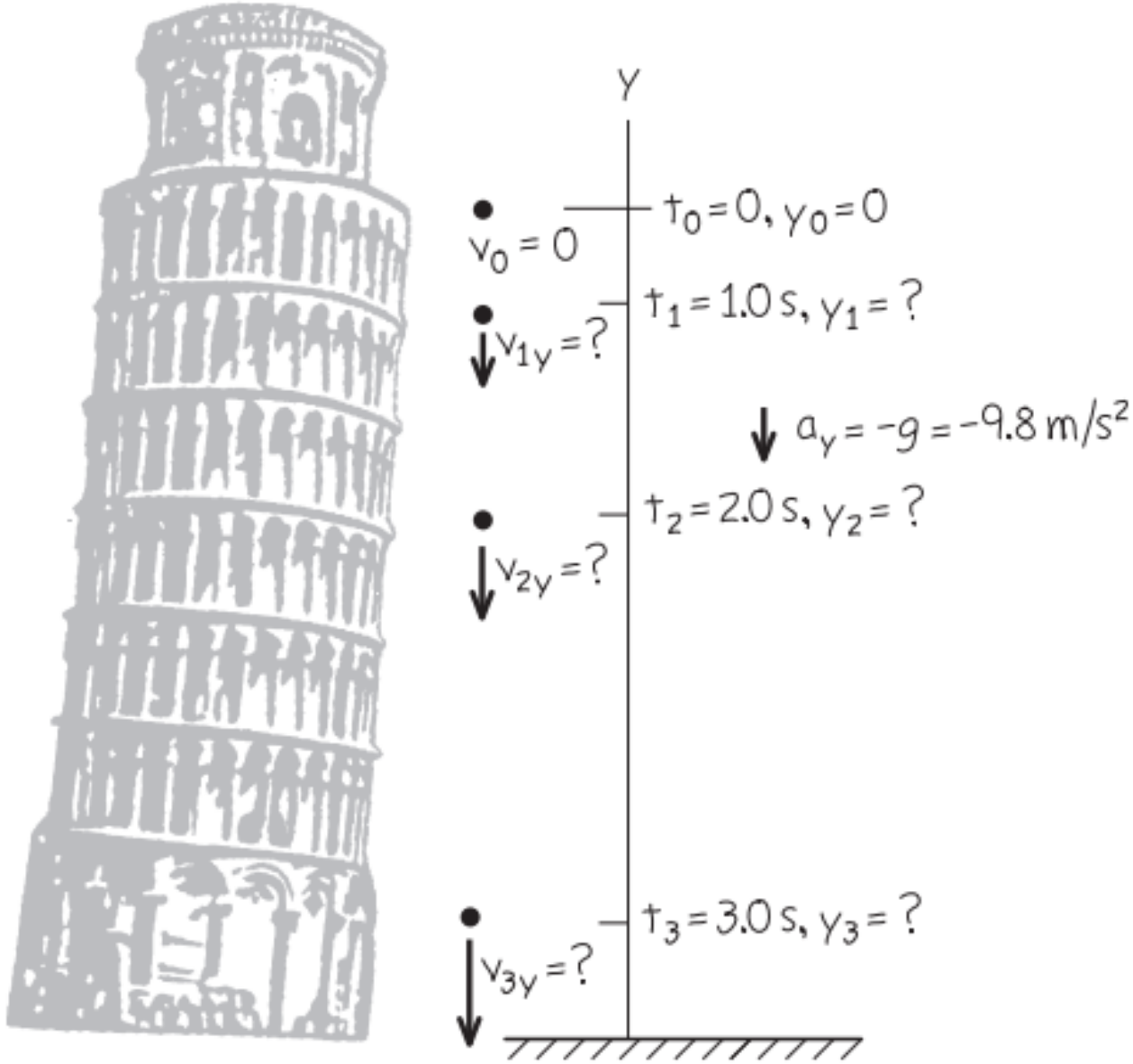
$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$(25)2 = (15)2 + 2(4)(x - 5)$$

$$x = 55\ m$$

Example

A one-euro coin is dropped from the Leaning Tower of Pisa and falls freely from rest. What are its position and velocity after 1.0 s, 2.0 s, and 3.0 s? Ignore air resistance.



Example

$$y = ?\,, t = 1,2,3\ s$$

$$v_y = ?\,, t = 1,2,3\ s$$

"Falls freely" means "falls with constant acceleration due to gravity," so we can use the constant-acceleration equations.
Vertical coordinate axis and the coordinate y instead of x.
Both the initial coordinate y_0 and initial y-velocity v_{0y} are zero.
The y-acceleration is downward (in the negative y-direction),
$a_y = -g = -9.8\ m/s^2$
(Remember that g is a positive quantity.)

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$v_y = v_{0y} + at$$

Example

$$y = ?\,, t = 1,2,3\ s$$

$$v_y = ?\,, t = 1,2,3\ s$$

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$y = 0 + (0)t + (1/2)(-g)t^2$$

$$y = (-4.9)t^2$$

$$v_y = v_{0y} + at$$

$$v_y = 0 + (-g)t$$

$$v_y = 0 + (-9.8)t$$

$$y = -4.9, -20, -40\ (m)$$

$$v_y = -9.8, -20, -29\ (m/s)$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\Delta v_x = a_x \Delta t$$

$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x dt$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

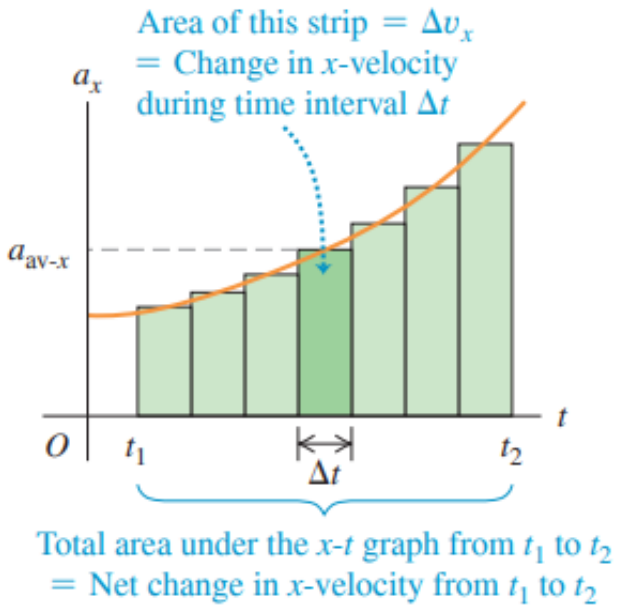
$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x dt$$

$$x_2 - x_1 = \int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v_x dt$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$



Example

A person is driving along a straight highway. At t=0, when she is moving at 10 m/s in the positive x-direction, she passes a signpost at x= 50 m. His x-acceleration as a function of time is

$$a_x = 2.0\ m/s^2 - (0.1\ m/s^3)t$$

(a) Find her x-velocity v_x and position x as functions of time.

(b) When is her x-velocity greatest?

(c) What is that maximum x-velocity?

(d) Where is the car when it reaches that maximum x-velocity?

Example

During your summer internship for an aerospace company, you are asked to design a small research rocket. The rocket is to be launched from rest from the earth's surface and is to reach a maximum height of 960 m above the earth's surface. The rocket's engines give the rocket an upward acceleration of 16.0 m/s^2 during the time T that they fire. After the engines shut off, the rocket is in free fall. Ignore air resistance.

What must be the value of T in order for the rocket to reach the required altitude?

Example

The rocket accelerates uniformly upward at 16.0 m/s2 with the engines on. After the engines are off, it moves upward but accelerates downward at 9.80 m/s^2.

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$v_y{}^2 = v_{0y}{}^2 + 2a_y(y - y_0)$$

both apply to both parts of the motion since the accelerations are both constant, but the accelerations are different in both cases. Let +y be upward.

Example

With the engines on, $v_{oy} = 0$, $a_y = 16.0$ m/s^2 upward, and t = T at the instant the engines just shut off. Using these quantities,

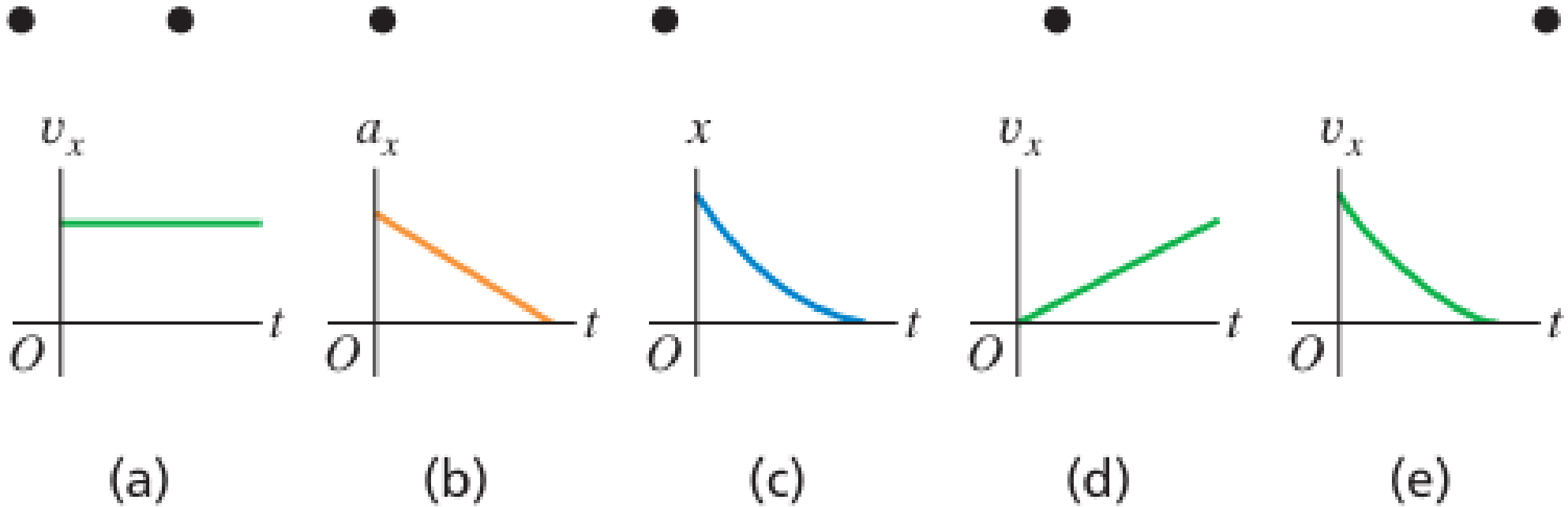
$$y = y_0 + v_{0y}t + (1/2)a_x t^2 = (1/2)(16)T^2$$

$$v_y{}^2 = v_{0y}{}^2 + 2ay(y - y_0) = (16)T$$

Technique	Model/Method	Suitable for
1. Analytical model	$x = x_0 + v_0 t + \frac{1}{2} a t^2$	Basic mechanics
2. Graphical model	$x - t,\ v - t,\ a - t$graphs	Teaching/visualization
3. Numerical modeling	Euler, Verlet, RK4	Computer simulation
4. Sensor-based model	Ultrasonic/IR/LiDAR sensor	Real-time measurement
5. Video analysis	Track an object's position frame-by-frame	Experimental physics
6. Encoder-based model	Rotary encoder → linear displacement	Robotics/mechatronics
7. Computer simulation	Python/MATLAB/Simulink	Simulation and analysis
8. AI/ML model	Measured t→ predicted x	Advanced modeling
9. Digital twin	Physical object + mathematical/computational model	Advanced laboratory
10. Interactive model	Slider/time input → moving object + graphs	Physics teaching

Example

The black dots at the top of figure represent a series of high speed photographs of an insect flying in a straight line from left to right (in the positive x-direction). Which of the graphs in figure most plausibly depicts this insect's motion?



Equations of Motion with Constant Acceleration

$$v_x = v_{0x} + at$$

$$x = x_0 + v_{0x}t + (1/2)axt^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = (1/2)(v_x + v_{0x})t$$

TABLE 2.5 Equations of Motion with Constant Acceleration

Equation		Includes Quantities			
$v_x = v_{0x} + a_x t$	(2.8)	t		v_x	a_x
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	(2.12)	t	x		a_x
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	(2.13)		x	v_x	a_x
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	(2.14)	t	x	v_x	

Position

A particle is at a point P at a certain instant, the position vector $\vec{r}$ of the particle at this instant is a vector that goes from the origin of the coordinate system to point P

The Cartesian coordinates x, y, and z of point P are the x-, y-, and z-components of vector $\vec{r}$.

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

